The Geometric Engine of Time: The Zero-Point Eight-Step Cycle, Euler's Formula in Hyperbolic Space, and a Conditional P = NP Equivalence

BioChain AI — SHD-CCP Architecture Research

Consolidated Technical Report (Continued Series)

Abstract

This installment of The Geometric Engine of Time explains the zero-point eight-step cycle, carries Euler's formula from the circle into hyperbolic space, and states exactly in what sense the Engine yields a conditional equivalence of P and NP. The cycle is Euler's formula made exact. The Gaussian integer 1 + i advances a spinor by 45° and a spatial rotation by 90°: four steps reach the spinor sheet −1 and eight return to +1. Niven's theorem explains why the clock needs projective integer arithmetic, since no rational unit quaternion has order 8. The nine phases of the cycle's seed split into three 120° triads offset by 40°, each the base of a regular tetrahedron of side 2√2 with a common apex. Lifted to a paraboloid, each tetrahedron and the origin span a 4-simplex, and in the conformal model R^{4,1} five null vectors turn squared distance into a bilinear form. A threefold Fourier split, a joint 24-frame schedule with a same-branch delay of 12, and the 720 → 72 → 10 pump complete the cycle. The zero point is a closure state tested three ways; the cyclic holonomy, identically 1, is not one of the tests. In hyperbolic space Euler's formula becomes e^{jϑ} = cosh ϑ + j sinh ϑ, whose unit group has only four elements of finite order: a hyperbolic motion never returns. Rotation angle and rapidity are the imaginary and real parts of one complex rapidity, and the Engine's three motions are its three cases. The spinor clock is elliptic. The scale register and the face-neighbour boosts of the {4,3,5} honeycomb are hyperbolic, with golden rapidity cosh ϑ = φ and a half-rapidity equal to the cell's inradius, 0.5306. The generator 1 + i is a loxodromic screw whose translation per step is (ln 2)/2, exactly the half bit by which its norm grows. Gauss–Bonnet and Euler's polyhedron formula fix the cell's curvature and give Euler characteristic 0 for every closed manifold tiled by these cubes. The complexity results rest on one theorem, the Euler evaluation dichotomy: an N-step power is evaluated exactly in constant time if and only if the motion has finite order; otherwise its exact value needs Θ(N) bits. The Engine's collapse P_GET = NP_GET is proven for problems whose witnesses are procedural evaluations, and is constant time only within bounded precision horizons, where every problem is. For classical complexity the exact statement is an equivalence: P = NP holds if and only if every problem in NP admits a polynomial-time integrable encoding of the kind the Engine's axioms

describe. Whether one exists is the P versus NP problem, which remains open; seven obstructions and the relativization barrier mark where the Engine's methods stop. The multi-scale predictions are recomputed, among them the cubic-shadow counts and Ramanujan mixing times of 4.20 against 269.30 on a cubic torus at about 110,000 nodes, and the errata of the three source texts are listed.

Editorial Note. This document is an installment of *The Geometric Engine of Time* (Hoard 2026a) and is cited in the series as Hoard (2026i). It is the companion to the monograph *Rigorous Formulation of Conditional P = NP, O(1) Hyperbolic Traversal Analysis, and Multi-Scale Predictions* (cited as Traversal Monograph) and to the research addendum *Zero-Point Eight-Step Pump and Three 120-Degree Conformal Lifts* (cited as Zero-Point Addendum). It also formalizes the text *The Volumetric Breath of the Zero-Point* (cited as Volumetric Breath). It abides by the proofs of *The Fifteen Lemmas of the Geometric Engine* (Hoard 2026, cited as Fifteen Lemmas), follows the conventions of *A Proposal for a Standardization of the Cubit* (Hoard 2026d), and uses the release paper on the Fokker-Planck prime sieve (Hoard 2026e) and its companion (Hoard 2026h). Where the Traversal Monograph states an obstruction for a particular application, this paper states it in its general form. All quantitative claims are annotated with status tags: **[PROVEN]**, **[HEURISTIC]**, **[CONJECTURAL]**, **[OPEN]**, **[SUPERSEDED]**, refined by five tags defined under Epistemic Conventions and Scope.

Contents

Epistemic Conventions and Scope 4
1 Sources 5
1.1 The Sources and Where They Enter 5
1.2 The Fifteen Lemmas in This Paper 6
1.3 Three Meanings of Dimension 7
2 Euler's Formula on the Circle: The Spinor Clock 7
2.1 Roots of Unity and the Nine-Phase Wheel 7
2.2 The Exact Generator 1 + i 8
2.3 Spinor Angle and Spatial Angle 8
2.4 Why the Clock Has Eight Steps: Niven's Bound 9
2.5 A Phase Lives on the Circle, Not the Line 9
3 The Eight-Step Cycle of the Zero-Point Topology 10
3.1 The Three Tetrahedra of the Nine-Phase Wheel 10
3.2 The Paraboloid and Conformal Lifts 11
3.3 The Threefold Decomposition 12
3.4 The Eight Stages 12
3.5 The Joint 24-Frame Schedule 13
3.6 Nesting in the 720 → 72 → 10 Pump 14
3.7 Recombination, Erasure and Folding 15
3.8 The Zero Point as a Closure State 15
4 Euler's Formula in Hyperbolic Space 16
4.1 The Hyperbolic Euler Formula 16
4.2 One Formula for Rotations and Boosts: Complex Rapidity 17
4.3 The Golden Rapidity of the {4,3,5} Honeycomb 17
4.4 The Eight-Step Clock as a Hyperbolic Screw 18
4.5 The Scale Register as a Boost 19
4.6 Euler's Polyhedron Formula and Gauss–Bonnet 19
4.7 Exponential Volume 20
4.8 The Clock and the Depth 21
5 Conditional P = NP: What the Cycle Makes Constant and What It Cannot 21
5.1 The Euler Evaluation Dichotomy 21
5.2 The Machine Model and the Classes 22
5.3 The Five Integrability Axioms 23
5.4 The Conditional Collapse 24
5.5 What the Collapse Is 24
5.6 The Conditional Equivalence 25
5.7 Verification and Search in the Hyperbolic Tree 26
5.8 The Boundary: Seven Obstructions 26
5.9 Barriers 27
5.10 The Constant-Time Pipeline 28
5.11 Where the Equivalence Holds 28
6 Multi-Scale Predictions 29
6.1 Hyperbolic Addresses Against the Cubic Shadow 29
6.2 Precision Horizons 30

6.3 Mixing on Ramanujan Graphs and Cubic Tori 30
6.4 Clock Windows of the Cycle 31
7 Relation to the Geometric Engine of Time 31
8 Status of Claims 33
9 Errata of the Source Texts 35
9.1 The Zero-Point Addendum 35
9.2 The Volumetric Breath 36
9.3 The Traversal Monograph 37
10 Open Problems 37
11 Conclusion 40
Appendix A: Computational Verification 40
Appendix B: Notation 41
References 42

Epistemic Conventions and Scope

Every claim carries one of ten tags: the nine of the series, as fixed in the cubit installment (Hoard 2026d, Table 1), and **[CORRESPONDENCE]**, adopted from the Fifteen Lemmas. A claim of a source text keeps the tag that the source's own argument supports, which is not always the tag the source gave itself.

Table 1: Status tags used in this installment.

Tag	Meaning	Example in this paper
[PROVEN]	A deductive result with a complete proof in this document or in a cited series document whose proof was re-checked.	Four steps of 1 + i give the spinor −1 and eight give +1 (Proposition 2.2)
[CORRESPONDENCE]	Two finite sets of equal size, or two equal numbers; no structure-preserving map is claimed.	lcm(8, 3) = 24 equals the order of the binary tetrahedral group (Section 3.5)
[DEFINITION]	A convention adopted here. It can be useful or not, but not true or false.	The joint frame register k = n mod 8, j = n mod 3 (Definition 3.6)
[ESTABLISHED]	Standard mathematics, cited and not re-proved.	Baker, Gill and Solovay: oracles exist on both sides of P versus NP
[COMPUTED]	A number obtained by evaluating a stated formula. Reproducible from Appendix A.	Mixing time 4.20 on X(5, 61)
[EMPIRICAL]	A comparison with measured data.	Not used in this paper
[HEURISTIC]	A modelling choice motivated by analogy, not derived.	The tetrahedral apex as a parity channel (Section 3.3)
[CONJECTURAL]	A claim that would need new evidence to be accepted.	A throughput gain from the 24-frame schedule (Problem 3)
[OPEN]	A stated question awaiting proof or test.	P versus NP
[SUPERSEDED]	A claim of a source replaced by a corrected statement.	"A perfect return to the identity at frame 720 purges drift" (Section 9.2)

Rule for mixed claims. As in the cubit installment, a computed value of a heuristic formula is tagged **[HEURISTIC / COMPUTED]**: the arithmetic is checked, but the modelling choice is not thereby established. The instruction budget of the constant-time pipeline (Section 5.10) is the example here.

What this paper does not claim.

- It does not prove P = NP, and it does not prove P ≠ NP. Section 5 proves an equivalence whose condition is exactly as hard as the problem itself.
- It does not claim constant-time computation for arbitrary problems. Every constant-time statement here holds either for a finite-order motion or inside a bounded precision horizon, and Section 5.5 shows why the second case says nothing asymptotic.
- It does not claim that a lift into five coordinates creates physical extra dimensions. The lifts are embeddings that change how many coordinates represent a point (Definition 1.1).
- It does not claim that the zero point is a physical singularity or a source of energy. It is defined as a closure state (Definition 3.12).
- It does not claim that symmetry makes arbitrary data compressible. Folding is lossless for symmetric tables, not for data (Proposition 3.11).

Relation to the series. Section 1 maps every source, and every lemma of the Fifteen Lemmas used here, to the sections that use it. Section 7 states where the Geometric Engine of Time supplies the most accurate method of calculation and where it is the source of a construction only.

Notation. Inline mathematics is set in plain text (for example q_k = e^{ikπ/4}); display equations are typeset. Appendix B collects the symbols.

1 Sources

The paper formalizes three source texts and draws on six further documents of the series. Every imported statement was re-checked before use (Appendix A).

1.1 The Sources and Where They Enter

Table 2: Where each source enters this paper.

Label	Document	What it supplies	Sections
Traversal Monograph	*The Geometric Engine of Time: Rigorous Formulation of Conditional P = NP, O(1) Hyperbolic Traversal Analysis, and Multi-Scale Predictions* (Hoard, October 2026)	The machine model, the classes SEQ(N), NP_GET and P_GET, the axioms C1 to C5, the collapse theorem, five obstructions, the pipeline budget, the multi-scale tables	2.2, 4.3, 5, 6
Zero-Point Addendum	*Zero-Point Eight-Step Pump and Three 120-Degree Conformal Lifts* (27 October 2026)	The three meanings of dimension, the nine-phase wheel and its tetrahedra, the paraboloid and conformal lifts, the C_3 decomposition, the 1 + i convention, the 24-frame schedule, recombination, the three closure tests	1.3, 3

Label	Document	What it supplies	Sections
Volumetric Breath	*The Volumetric Breath of the Zero-Point: An 8-Step π/4 Guide to Cubic Expansion, Hyperbolic Projection, and Symmetry Folding* (17 October 2026)	The eight named stages of the cycle	3.4, 9.2
Twelve-Engine Memory	*Memory in the Twelve-Engine Hierarchy* (16 October 2026), §§4–6	The 180° handshake apices as distinct from the 120° tier rotation; the 24-step schedule as an open task	3.5
Hyperbolic Horizon	*Navigating the Hyperbolic Horizon: Achieving O(1) Traversal in O(N) Sequential Systems using 64-bit Quaternion Cubes*	Theorem 3, the three nested lookups	5.4, 9.3
Fifteen Lemmas	*The Fifteen Lemmas of the Geometric Engine* (Hoard 2026)	Proofs of Lemmas 1 to 15; the Einstein Tile	2 to 6
Hoard 2026a	*Mathematical Architecture of the 6-Channel, 5-Tier Toroidal Orbital Calculator*	$W_\Sigma = 60$, the stator tier laws, the $C_3 \times C_2$ channels, the 30-pulse closure	5.2, 6.4
Hoard 2026b	*The Torsional Markov Memory-Offload Pump*	The 720 → 72 → 10 gears and the block partition	3.6
Hoard 2026d	*A Proposal for a Standardization of the Cubit*	Tag conventions; the cycle-current formula; the vacuous cyclic holonomy; the cubit's uncertainty	3.8, 5.8, 6.2
Hoard 2026e and 2026h	*The Fokker-Planck Prime Sieve in Quaternionic Lattices*, release and companion	The norm-5 tree and its unique factorization, Ramanujan quotients, precision horizons, the Circularity Boundary, the trefoil folding correction	4.7, 5.7, 5.8, 6

1.2 The Fifteen Lemmas in This Paper

Eleven of the fifteen lemmas carry results this paper uses, listed in Table 3. Lemmas 1 to 3 enter only through Lemma 5, and Lemma 10 (the disjoint tier shells) only through the machine model of Section 5.2.

Table 3: Results of the Fifteen Lemmas used in this paper.

Fifteen Lemmas result	Used in	What it does here
Lemma 4, Part 4	5.3	The tier spinor $h = (1 - i + j + k)/2$ has order 6 and $h\,h\,\bar{h}\,\bar{h} = 1$
Lemma 5, Part 2	2.4, 4.8	The cube group O and its spin lift 2O
Lemma 6, Parts 1 to 4	5.3	Well-conditioning of the five-moment Vandermonde system (axiom C4), cited only
Lemma 7, Part 3(d) and (e)	3.7, 5.8	Four records closed by a product survive one loss, not two; drift without a potential does not select
Lemma 8, Steps 2 and 6; Propositions 6.2 to 6.5, 7.16, 7.17, 8.2″	3.1, 3.2, 3.7, 4.3	The circle-to-sphere lift and its nine rays; the conformal lift; the ghost map; the {4,3,5} boosts; the inclusion $SU(2) \subset SL(2, \mathbb{C})$; the trefoil's D_3
Lemma 9, Parts 2 and 4	5.8, 6.2	Closed-form CRT addresses; the pigeonhole bound and the triviality of maps from finite groups into torsion-free groups
Lemma 11, Proposition 11.4	3.5, 6.4	The clock windows of 2, 6, 10 and 30 pulses
Lemma 12, Parts 1 to 3	3.6, 3.8	Block telescoping; the vacuous cyclic holonomy; prefix-scan evaluation in O(log k) depth

Fifteen Lemmas result	Used in	What it does here
Lemma 13, Parts 1 to 4	2.4, 5.1	Rational finite orders are 1, 2, 3, 4 or 6; denominators grow without bound; integer bit growth; gcd reduction of (1 + i)^k
Lemma 14, Part 4; Lemma 15	5.8	The Circularity Boundary; an 8-bit header filters but does not certify

1.3 Three Meanings of Dimension

The sources speak of lifts from one to two to five dimensions. The word has three meanings, and the paper keeps them apart.

> **Definition 1.1 (Dimension).** **[DEFINITION]** Zero-Point Addendum, §1.
>
> *The parameter dimension of a set of points is the number of independent inputs that determine a point. The ambient dimension is the number of coordinates used to represent a point. The span dimension of a finite configuration is the dimension of the vector space its points generate together.*

A circle has parameter dimension 1 and, in the plane, ambient dimension 2. A regular tetrahedron spans a three-dimensional affine space. Its conformal image uses five coordinates, and five of its lifted points span R^5 (Proposition 3.3). A map from a one- or two-parameter domain does not acquire five intrinsic degrees of freedom: the phrase "2D-to-5D lift" means a sequence of embeddings that raises the ambient dimension, with a five-dimensional span reached by a whole configuration, never by a single branch.

2 Euler's Formula on the Circle: The Spinor Clock

Status: **[PROVEN]** throughout. Sources: Traversal Monograph §5.1; Zero-Point Addendum §§2, 5; Fifteen Lemmas, Lemmas 5, 9 and 13.

The eight-step cycle is Euler's formula evaluated at multiples of 45° in exact arithmetic. This section derives the cycle from the formula, shows why it has eight steps and not six or twelve, and separates the spinor angle from the spatial angle that the sources sometimes confuse.

2.1 Roots of Unity and the Nine-Phase Wheel

Euler's formula sends an angle to a point of the unit circle:

$$e^{i\theta} = \cos\theta + i\sin\theta, \qquad e^{i\pi} + 1 = 0.$$

Its values at the n-th parts of a turn are the n-th roots of unity μ_n = {e^{2πis/n}}, a cyclic group of order n. The seed of the cycle is the wheel μ_9 of nine phases.

> **Proposition 2.1 (The Nine-Phase Factorization).** **[PROVEN]** Zero-Point Addendum, §2; Fifteen Lemmas, Lemma 8.
>
> *Every s in $\mathbb{Z}/9$ is uniquely s = a + 3j with a, j ∈ {0, 1, 2}, so*
>
> $$\theta_{a,j} = \frac{2\pi a}{9} + \frac{2\pi j}{3}, \qquad \mu_9 = \bigsqcup_{a=0}^{2} \zeta_9^{\,a}\,\mu_3, \qquad \zeta_9 = e^{2\pi i/9}.$$
>
> *The nine phases are three 120° triads, the cosets of μ_3, offset from one another by 40°, and each triad sums to zero.*

Proof. Division with remainder by 3 gives the unique decomposition, and e^{2πi(a+3j)/9} = ζ₉^a ω^j with ω = e^{2πi/3}. The triad of index a is ζ₉^a μ₃, and its sum is ζ₉^a(1 + ω + ω²) = 0 because ω is a root of x² + x + 1. ■

Two threefold indices appear and must not be merged: a labels a family (a tetrahedron, Section 3.1) and j labels a branch within it. The triad sum is the case n = 3 of the identity Σ e^{2πis/n} = 0, whose case n = 2 is e^{iπ} + 1 = 0.

2.2 The Exact Generator 1 + i

The Engine runs the cycle on the Gaussian integer 1 + i rather than on the unit number e^{iπ/4}, because 1 + i has integer coordinates. Euler's formula separates its size from its angle:

$$1 + i = \sqrt{2}\, e^{i\pi/4} = e^{\frac{\ln 2}{2} + i\frac{\pi}{4}}, \qquad q_k = \frac{(1+i)^k}{|1+i|^k} = e^{ik\pi/4}.$$

Proposition 2.2 (Powers of the Generator). **[PROVEN]** Fifteen Lemmas, Lemma 13 Part 4; Traversal Monograph §5.1.

(1) (1 + i)² = 2i, (1 + i)⁴ = −4 and (1 + i)⁸ = 16.

(2) Dividing (1 + i)^k by the greatest common divisor of its coordinates gives, for k = 1 to 8, the states 1 + i, i, −1 + i, −1, −1 − i, −i, 1 − i, 1. Each coordinate lies in {−1, 0, 1}, so a state needs two signed bits per coordinate and the step index needs three bits.

(3) The normalized states are q_k = e^{ikπ/4}: q₄ = −1 and q₈ = +1.

Proof. (1 + i)² = 1 + 2i + i² = 2i, so (1 + i)⁴ = (2i)² = −4 and (1 + i)⁸ = 16. Write (1 + i)^k = 2^{⌊k/2⌋} · ε_k with ε_k ∈ {1 + i, i, −1 + i, −1, ...} the reduced state; the gcd of the coordinates of (1 + i)^k is 2^{⌊k/2⌋} because 1 + i itself has coprime coordinates and i^m(1 + i) does too. Part (3) is Euler's formula applied to the polar form above. ■

2.3 Spinor Angle and Spatial Angle

A unit quaternion q = cos α + u sin α, with u a unit pure quaternion, rotates three-space by x ↦ qxq̄, through the angle 2α about the axis u **[ESTABLISHED]**. The parameter α is the spinor angle; 2α is the spatial angle. Every statement about the cycle must say which one it means.

Proposition 2.3 (The Cycle in Space and in Spin). **[PROVEN]** Zero-Point Addendum, §5; Fifteen Lemmas, Lemma 8 and Theorem 7.5.

Let q_k = e^{ikπ/4}, read as the quaternion cos(kπ/4) + i sin(kπ/4).

(1) The map x ↦ (1 + i)^k x (1 − i)^k / 2^k is the integer rotation Rot_x(90k°): each step turns space by 90° about the x-axis, and four steps give the identity rotation.

(2) Steps 2 and 6 give the same half-turn diag(1, −1, −1) with opposite spinors +i and −i. Step 4 gives the identity rotation with spinor −1; only Step 8 returns the spinor to +1. In SU(2), four steps give −I and eight give +I.

(3) The transverse envelope A_k = |sin(kπ/4)| equals 1 exactly at k = 2 and k = 6, and 0 exactly at k = 0, 4 and 8. The two handshake windows are 180° apart in spinor

angle.

Proof. With $\alpha = k\pi/4$ the spatial angle is $k\pi/2 = 90k°$, and the conjugation formula, computed exactly, gives the integer matrix Rot_x(90°)^k. Rot_x(180°) = diag(1, −1, −1), and $q_6 = -q_2$ because $q_4 = -1$; a quaternion and its negative give the same rotation. In SU(2) the step is $\exp(-i(\pi/2)\sigma_1/2)$, with trace $2\cos(\pi/4) = \sqrt{2}$, and its fourth power is $\exp(-i\pi\sigma_1) = -I$. Part (3) is the table of $|\sin(k\pi/4)|$. ■

Two engines that meet at Steps 2 and 6 see the same spatial geometry without agreeing on the sheet of the double cover, which is why the Engine places its onload and offload handshakes there (Fifteen Lemmas, Theorem 7.5). Step 4 is the opposite of an apex: the envelope vanishes there.

2.4 Why the Clock Has Eight Steps: Niven's Bound

Proposition 2.4 (No Rational Clock of Order Eight). **[PROVEN]** Fifteen Lemmas, Lemma 13 Part 1 and Lemma 5 Part 2; Niven (1956).

(1) A rational unit quaternion of finite order has order 1, 2, 3, 4 or 6.

(2) No rational unit quaternion has order 8. The spinor step $(1 + i)/\sqrt{2}$ has irrational coordinates.

(3) The cycle $\langle(1 + i)/\sqrt{2}\rangle \cong \mathbb{Z}/8$ lies in the binary octahedral group 2O of order 48, the spin lift of the cube group O, and not in the binary tetrahedral group 2T of the 24 Hurwitz units; it meets 2T in $\{\pm 1, \pm i\}$. Its image in SO(3) is Rot_x(90°) ∈ O, an integer matrix.

Proof. (1) If $q = \cos\alpha + u\sin\alpha$ has order n, then $\alpha = 2\pi m/n$ in lowest terms and $\cos\alpha$ is rational. By Niven's theorem the rational values of the cosine at rational multiples of π are 0, ±1/2 and ±1, which occur exactly for $n \in \{1, 2, 3, 4, 6\}$. (2) Order 8 would need $\cos\alpha = \pm\sqrt{2}/2$. (3) Conjugation by $(1 + i)/\sqrt{2}$ is the 90° rotation about the x-axis, which permutes the coordinate axes and so normalizes 2T; adjoining it to 2T gives 2O. The elements of order 8 in 2O are the twelve $(\pm 1 \pm u)/\sqrt{2}$ with $u \in \{i, j, k\}$, none of them in 2T. ■

So the spatial clock is rational, since Rot_x(90°) is an integer matrix, but its spinor lift is not. The Engine keeps exactness by running the clock projectively: it multiplies by the integer 1 + i and removes the common factor, which Proposition 2.2 bounds by two signed bits. Appendix A, section C, checks part (1) on all 39,624 rational unit quaternions of denominator at most 25.

2.5 A Phase Lives on the Circle, Not the Line

Proposition 2.5 (Circle-Valued and Line-Valued Phases). **[PROVEN]** Fifteen Lemmas, Lemma 9 Part 4(b).

The character $k \mapsto e^{ik\pi/4}$ is a faithful homomorphism $\mathbb{Z}/8 \to S^1$. Every homomorphism from $\mathbb{Z}/8$, or from any finite group, into a torsion-free group such as $(\mathbb{Q}^d, +)$ is zero.

Proof. The eight values $e^{ik\pi/4}$ are distinct and multiply by adding exponents mod 8. If φ is a homomorphism from a group of order n into a torsion-free group, $n\varphi(g) =$

φ(gⁿ) = 0 for every g, so φ(g) = 0. ■

Euler's map t ↦ e^{it} is the wrap that turns the line into the circle: its kernel is 2π$\mathbb{Z}$, and that kernel is what lets a phase return. A cycle position therefore cannot be stored as any linear coordinate; it is a residue, held by the Einstein Tile as the three-bit step field of the South Halo. Section 4.1 shows that the hyperbolic form of Euler's map has no kernel, which is the root of every obstruction in Section 5.

3 The Eight-Step Cycle of the Zero-Point Topology

Status: **[PROVEN]**, with the schedule a **[DEFINITION]** and one interpretation **[HEURISTIC]**. Sources: Zero-Point Addendum §§2–9; Volumetric Breath; Fifteen Lemmas, Lemmas 7, 8 and 12; Hoard 2026b.

The cycle has a geometric seed and a temporal clock. The seed is the nine-phase wheel of Section 2.1, which carries three regular tetrahedra and lifts to a 4-simplex and to five null vectors. The clock is the spinor cycle of Section 2.3. A joint register of period 24 binds them, and the 720 → 72 → 10 pump nests them. The zero point is the state in which all of this closes.

3.1 The Three Tetrahedra of the Nine-Phase Wheel

Let u, v be orthonormal vectors perpendicular to (1, 1, 1), and set c = −(1, 1, 1)/3, r = √(8/3) and x* = (1, 1, 1). Place the wheel on the circle

$$x_{a,j} = c + r\left(\cos\theta_{a,j}\, u + \sin\theta_{a,j}\, v\right), \qquad \Delta_a = \{x_{a,0}, x_{a,1}, x_{a,2}, x_*\}.$$

Proposition 3.1 (Three Tetrahedra). **[PROVEN]** Zero-Point Addendum, §3; Fifteen Lemmas, Lemma 8, Proposition 8.2″; frame of part (4) corrected here.

(1) Every x_{a,j} lies on the sphere ‖x‖² = 3, the circumsphere of the core cube K = [−1, 1]³, and x_{a,0} + x_{a,1} + x_{a,2} = 3c = −x.*

(2) Each Δ_a is a regular tetrahedron of side 2√2: all six squared distances among its vertices equal 8.

(3) The rotation by 40° about the axis (1, 1, 1) carries Δ₀ to Δ₁ to Δ₂ to Δ₀. It is not in the cube group O, since cos 40° is irrational.

(4) The circle contains exactly three vertices of K, namely (1, −1, −1), (−1, 1, −1) and (−1, −1, 1), 120° apart. In the frame u = (2, −1, −1)/√6, v = (0, 1, −1)/√2 they are the triad a = 0, and no other point of the wheel is a vertex of K. Then Δ₀ is the tetrahedron inscribed in K, and Δ₀ ∪ (−Δ₀) is the set of all eight vertices.

Proof. (1) c is perpendicular to u and v, so ‖x‖² = ‖c‖² + r² = 1/3 + 8/3 = 3; the three unit vectors at 120° sum to zero, leaving 3c. (2) Two branch points 120° apart are at squared distance 2r²(1 − cos 120°) = 3r² = 8; and x − x = 4c + r(...) has squared length 16/3 + 8/3 = 8. (3) The rotation fixes c and x* and turns the plane of u and v by 40°, so it adds 40° to every phase, and θ + 120° returns the third family to the first. The minimal polynomial of cos 40° has degree 3, while O consists of signed permutation matrices. (4) Every vertex of K has ‖x‖² = 3, and the circle is the intersection of that sphere with the plane x·(1, 1, 1) = −1, so its vertices are those with coordinate sum −1. With the stated u, v, the phase 0 gives c + ru = (1, −1, −1)*

and the phase 120° gives (−1, 1, −1). The cube vertices sit at multiples of 120°, which are wheel phases only for a = 0. ■

Correction (The Frame). The Zero-Point Addendum takes u = (1, −1, 0)/√2 and v = (1, 1, −2)/√6 and states that the triad a = 0 lies on cube vertices. In that frame the three cube vertices of the circle sit at 30°, 150° and 270°, none of which is a multiple of 40°, so no point of the wheel is a cube vertex **[SUPERSEDED]**. Parts (1) to (3) do not depend on the frame; part (4) holds in the frame given above, which is the frame in which the Fifteen Lemmas' statement that "exactly 3 of 9 rays end at cube vertices" is true. Appendix A, section E, checks both frames.

3.2 The Paraboloid and Conformal Lifts

Each branch point is lifted twice: to the paraboloid P(x) = (x, ‖x‖²) in R⁴ and to the conformal null cone X(x) = (x, ‖x‖²/2, 1) in R^{4,1}, with the bilinear form

$$\langle (x,\alpha,\beta),(y,\alpha',\beta')\rangle = x\cdot y - \alpha\beta' - \beta\alpha'.$$

In the notation of the Traversal Monograph, X(x) = x + ½‖x‖²e_∞ + e₀, with e_∞ the α-direction and e₀ = X(0).

Proposition 3.2 (The 4-Simplex). **[PROVEN]** Zero-Point Addendum, §3.1; Fifteen Lemmas, Lemma 8.3.

For each family a, the five points P(0), P(x), P(x_{a,0}), P(x_{a,1}), P(x_{a,2}) are affinely independent: they span a non-degenerate 4-simplex in R⁴.*

Proof. The differences from P(0) are the rows (x, 3) for the four vertices x of Δ_a. If Σ c_i(x_i, 3) = 0, then Σ c_i x_i = 0 and Σ c_i = 0; the vertices of a non-degenerate tetrahedron are affinely independent, so every c_i = 0, and the four rows have rank 4. ■

Proposition 3.3 (The Conformal Null Cone). **[PROVEN]** Zero-Point Addendum, §3.2; Fifteen Lemmas, Lemma 8 Step 2.1 and Proposition 7.17.

(1) The form has signature (4, 1).

(2) ⟨X(x), X(x)⟩ = 0 and −2⟨X(x), X(y)⟩ = ‖x − y‖² for all x, y.

(3) For each family a, the five null vectors X(0), X(x), X(x_{a,0}), X(x_{a,1}), X(x_{a,2}) span R⁵.*

(4) The sphere of centre m and radius ρ is the set of x with ⟨X(x), S⟩ = 0, where S = X(m) − (ρ²/2)e_∞ and ⟨S, S⟩ = ρ². The circumsphere of K is S_K = e₀ − (3/2)e_∞.

Proof. (1) The block of α and β is [[0, −1], [−1, 0]], with eigenvalues ±1. (2) ⟨X(x), X(y)⟩ = x·y − ‖x‖²/2 − ‖y‖²/2 = −‖x − y‖²/2. (3) Subtracting X(0) = (0, 0, 1) leaves the rows (x, 3/2, 0), of rank 4 by Proposition 3.2, and X(0) supplies the last coordinate. (4) ⟨X(x), e_∞⟩ = −1, so ⟨X(x), S⟩ = (ρ² − ‖x − m‖²)/2; and ⟨S, S⟩ = 0 + ρ² because ⟨X(m), e_∞⟩ = −1 and e_∞ is null. Take m = 0, ρ² = 3. ■

Part (2) is the linearization of axiom C2 (Section 5.3): for a fixed point x, the quadratic condition ‖x − y‖² = d becomes the linear condition ⟨X(x), X(y)⟩ = −d/2 on the null vector X(y), and part (4) does the same for spheres. All three branches use the same R^{4,1} with

different phase labels; they are not three independent five-dimensional spaces, and a single branch, a curve, never spans more than its own few dimensions (Definition 1.1).

3.3 The Threefold Decomposition

Proposition 3.4 (The C_3 Fourier Split). **[PROVEN]** Zero-Point Addendum, §4.

For three real channels (a, b, c) let s = a + b + c and z = a + ωb + ω²c, with ω = e^{2πi/3}. Then

$$a = \tfrac{1}{3}\left(s + z + \bar{z}\right), \qquad b = \tfrac{1}{3}\left(s + \omega^2 z + \omega \bar{z}\right), \qquad c = \tfrac{1}{3}\left(s + \omega z + \omega^2 \bar{z}\right).$$

The cyclic shift (a, b, c) ↦ (c, a, b) fixes s and multiplies z by ω, so $R^3 \cong R \oplus R^2$ as a representation of C_3: one fixed line and one rotating plane. Three equal-weight phasors on one complex carrier carry only z, two real numbers, so they cannot carry three independent real channels; three time slots can.

Proof. Substitute and use 1 + ω + ω² = 0; the shifted transform is c + ωa + ω²b = ω(a + ωb + ω²c). ■

Remark (The Apex as a Parity Channel). **[HEURISTIC]** The fixed apex x* of the three tetrahedra and the fixed scalar s play the same role, an invariant beside a rotating part, which suggests storing a parity or reference value at the apex. The analogy does not store anything; it needs an explicit encoder and decoder (Problem 1).

3.4 The Eight Stages

Definition 3.5 (The Eight-Step Cycle). **[DEFINITION]** Volumetric Breath; Zero-Point Addendum, §5.

The cycle is the sequence of states q_k = e^{ikπ/4}, k = 0, …, 8, held exactly as the reduced Gaussian states of Proposition 2.2, with the stages of Table 4. Each stage names the operation the Engine performs while the clock is in that state.

Table 4: The eight stages. Spatial angle is about the x-axis; A_k = |sin(kπ/4)|.

k	Stage	Spinor q_k	Reduced state	Spatial angle	A_k	Operation	Status
0	Zero point	+1	1	0°	0	Reference state; closure tests of Section 3.8	**[DEFINITION]**
1	Seed	(1 + i)/√2	1 + i	90°	0.707	Launch the nine-phase wheel and its three tetrahedra (3.1)	**[PROVEN]** geometry
2	Onload	+i	i	180°	1	Lift the branch to the 4-simplex and the null cone (3.2); first handshake	**[PROVEN]**
3	Project	(−1 + i)/√2	−1 + i	270°	0.707	Map into hyperbolic space by the ghost map or a face-neighbour boost (4.3)	**[PROVEN]** maps; their use **[HEURISTIC]**
4	Invert	−1	−1	360°	0	Spatial identity on the negative spinor sheet; a node of the envelope, not an apex	**[PROVEN]**
5	Fold	(−1 − i)/√2	−1 − i	450°	0.707	Apply the inverse transforms in the reverse order	**[PROVEN]** when the inverses are exact

k	Stage	Spinor q_k	Reduced state	Spatial angle	A_k	Operation	Status
6	Offload	$-i$	$-i$	540°	1	Pull back and recombine the branches (3.7); second handshake	**[PROVEN]** identity
7	Index	$(1-i)/\sqrt{2}$	$1-i$	630°	0.707	Rebuild symmetric tables from a fundamental domain; look up	**[PROVEN]** for symmetric tables
8	Snap	+1	1	720°	0	Spinor identity; record the holonomy residual	**[PROVEN]**

The spatial angle reaches 360° at Step 4 and 720° at Step 8, so one cycle is the double turn of the belt trick, completed in eight steps of $1 + i$ (Proposition 2.3).

3.5 The Joint 24-Frame Schedule

Definition 3.6 (Joint Register). **[DEFINITION]** Zero-Point Addendum, §6.

Advance the clock and the branch together once per frame: $k(n) = n \bmod 8$ and $j(n) = n \bmod 3$. The branch j is onloaded at frames with $k = 2$ and offloaded at frames with $k = 6$.

Proposition 3.7 (The Schedule). **[PROVEN]** under Definition 3.6.

(1) $n \mapsto (n \bmod 8, n \bmod 3)$ is an isomorphism $\mathbb{Z}/24 \cong \mathbb{Z}/8 \times \mathbb{Z}/3$, so the joint state has period $\mathrm{lcm}(8, 3) = 24$.

(2) The onloads fall at frames 2, 10, 18 on branches 2, 1, 0, and the offloads at frames 6, 14, 22 on branches 0, 2, 1 (Table 5): each branch is onloaded once and offloaded once per 24 frames.

(3) A branch reaches its offload 12 frames after its onload. Twelve is the unique solution modulo 24 of $\Delta \equiv 4 \pmod 8$ and $\Delta \equiv 0 \pmod 3$.

(4) $720 = 30 \times 24 = 90 \times 8$: a 720-frame register holds 30 joint cycles and 90 spinor cycles, with 30 onloads and 30 offloads on each branch.

Proof. (1) is the Chinese remainder theorem, since $\gcd(8, 3) = 1$. (2) lists the $n < 24$ with $n \equiv 2$ or $6 \pmod 8$. (3) The offload must be 4 steps of the clock later, modulo 8, on the same branch, so $\Delta \equiv 0 \pmod 3$; by (1) the solution is unique mod 24, and it is 12. (4) is arithmetic. ■

Table 5: One joint cycle of 24 frames. Branch angle $120j°$.

Frame n	Clock k	Branch j	Branch angle	Action
2	2	2	240°	Onload
6	6	0	0°	Offload of the branch onloaded at frame 18 of the previous cycle
10	2	1	120°	Onload
14	6	2	240°	Offload of frame 2
18	2	0	0°	Onload
22	6	1	120°	Offload of frame 10

The handshake apices are 180° apart in spinor angle, the branches 120° apart in phase; the schedule is the exact interlock of these two different symmetries, which the Twelve-Engine Memory note left as an open scheduling task. It is a clocking convention, not a theorem that any transport takes 12 frames.

Proposition 3.8 (Clock Windows of the Schedule). **[PROVEN]** with the frame a **[DEFINITION]**; Fifteen Lemmas, Lemma 11, Proposition 11.4.

Take one frame to be one pulse of 1/60 s. The caesium-133 frequency 9,192,631,770 Hz completes an integer number of periods after an even number of frames, the 40 Hz carrier after a multiple of 3 and the 432 Hz carrier after a multiple of 5. The joint schedule and all three clocks first close together after lcm(24, 2, 3, 5) = 120 frames, or 2 s, and the 720-frame register contains six such closures.

Proof. ν_Cs/60 = 153,210,529.5, so n frames hold an integer number of caesium periods exactly when n is even; 40n/60 = 2n/3 and 432n/60 = 36n/5 are integers exactly when 3 | n and 5 | n. ■

Remark (Correspondences). **[CORRESPONDENCE]** The period 24 equals the order of the binary tetrahedral group 2T and of the cube group O; the closure 120 equals the order of the binary icosahedral group 2I; the winding total $W_\Sigma = 60$ equals the order of the icosahedral rotation group. These are equal numbers. No homomorphism from $\mathbb{Z}/24$ onto 2T exists, since 2T is not abelian.

3.6 Nesting in the 720 → 72 → 10 Pump

The pump of Hoard (2026b) has an outer 12-point gear of 720 fine frames $q_0, \ldots, q_{719}$, a middle 10-point gear of 72 inverse aggregates, and an inner 8-point gear of 10 coarse frames. The inner gear's eight points are not the eight steps of the clock; equal counts do not identify the objects.

Proposition 3.9 (Block Telescoping and the Pump Holonomy). **[PROVEN]** Hoard 2026b; Fifteen Lemmas, Lemma 12; Zero-Point Addendum, §8.

Let $\Delta q_n = q_n\, q_{n-1}^{-1}$ with the open boundary $q_{-1} = q_0$, and let $P = \Delta q_{719} \cdots \Delta q_1 = q_{719}\, q_0^{-1}$.

(1) Each ten-frame block telescopes: $\Delta Q_m = \Delta q_{10m+9} \cdots \Delta q_{10m} = q_{10m+9}\, q_{10m-1}^{-1}$.

(2) With $\delta_m = \Delta Q_m^{-1}$ and the blocks $A_\ell = \{m : \lfloor 10m/72 \rfloor = \ell\}$, of sizes (8, 7, 7, 7, 7, 8, 7, 7, 7, 7), the right-acting products satisfy $Q^{(10)} = \delta_0\delta_1 \cdots \delta_{71} = P^{-1}$ and $Q^{(8)} =$ (product over ℓ of the products of δ_m over m in A_ℓ) $= P^{-1}$, and the holonomy is

$$H = Q^{(8)}\, Q^{(10)}\, P = P^{-1}.$$

(3) The cycle closes, $H = 1$, if and only if $P = 1$. Telescoping preserves the net boundary rotation and nothing else: distinct histories with the same endpoints give the same H.

Proof. (1) Adjacent factors cancel, $q_n\, q_{n-1}^{-1}\, q_{n-1}\, q_{n-2}^{-1} = q_n\, q_{n-2}^{-1}$. (2) $(AB)^{-1} = B^{-1}A^{-1}$ gives $\delta_0 \cdots \delta_{71} = (\Delta Q_{71} \cdots$

$\Delta Q_0)^{-1} = P^{-1}$; the A_ℓ are consecutive runs, so regrouping does not change the product; the sizes follow from [10m/72]. (3) follows from (2). ■

3.7 Recombination, Erasure and Folding

At the offload window the three branches return to the core. Let μ_j be the data of branch j after the lift and B_j an exactly invertible map to its target cell, and define $\mu_{core} = (1/3)\Sigma_j (B_j^{-1})_*\mu_j$ (Fifteen Lemmas, Proposition 7.17).

Proposition 3.10 (Single Erasure, Not History). **[PROVEN]** Zero-Point Addendum, §7; Fifteen Lemmas, Lemma 7 Part 3(d) and §7.2.

(1) If μ_{core} and two aligned branches are known, the third is recovered exactly: $(B_2^{-1})_*\mu_2 = 3\mu_{core} - (B_0^{-1})_*\mu_0 - (B_1^{-1})_*\mu_1$.

(2) The average alone does not determine the branches: for data in R^d the averaging map $R^{3d} \to R^d$ has a kernel of dimension 2d.

(3) Likewise, four records closed by $H_1H_2H_3H_4 = 1$ survive the loss of any one, H_a being the inverse product of the other three in cyclic order, but not of two: $H_3 = X$, $H_4 = X^{-1}(H_1H_2)^{-1}$ closes for every X. And the $24^3 = 13{,}824$ three-step chains of Hurwitz units give only 24 products, 576 chains each.

Proof. (1) rearranges the definition; (2) is rank-nullity, since the averaging map is onto. (3) is direct substitution, and the count holds because left multiplication by a fixed unit permutes the 24 units, so each product is reached $24^2 = 576$ times. ■

Proposition 3.11 (What Folding Saves). **[PROVEN]** Fifteen Lemmas, Lemma 8 Step 6; Hoard 2026h, Proposition 4.1.

The trefoil $T(t) = (\sin t + 2 \sin 2t, \cos t - 2 \cos 2t, -s \sin 3t)$ has the dihedral symmetry D_3 of order 6: $T(t + 2\pi/3)$ is $T(t)$ turned by 120° about the z-axis, and $T(-t) = \mathrm{diag}(-1, 1, -1)T(t)$. The arc $t \in [0, \pi/3]$ and its six images rebuild the curve, so a table of the curve shrinks to one sixth, and to one third under C_3 alone. A map from six arcs to one is injective only on D_3-symmetric inputs, so data written along the curve folds without loss only when the data is itself symmetric.

Proof. Expand sin and cos at $t + 2\pi/3$ and at $-t$; the six group images of the arc cover $[0, 2\pi)$. The last statement is the pigeonhole principle on six arcs. ■

With the knot's axis turned to (1, 1, 1), the symmetry is generated by the signed permutation matrices g_3 and g_2 of the Traversal Monograph, a D_3 inside O, so the reconstruction uses integer matrices only (Appendix A, section K).

3.8 The Zero Point as a Closure State

Definition 3.12 (The Zero Point). **[DEFINITION]** Zero-Point Addendum, §9.

The zero point is the state in which the chosen orientation, metric and memory invariants have returned to their prescribed values. It is checked by three separate tests: the spinor test, $q_8 = +1$ and not merely $q_4 = -1$; the geometric test, that every branch transform followed by its exact inverse restores the metric, the tetrahedral edge lengths and the core cube; and the memory test, that the open holonomy P =

$q_{719}\, q_0^{-1}$ equals 1, or that a recorded residual $P \neq 1$ is removed by a specified compensator.

The zero point is not a point of infinite density and not a physical singularity; it is a reference state with a test attached.

Proposition 3.13 (The Cyclic Holonomy Is No Test). **[PROVEN]** Hoard 2026d, Proposition 4.6; Fifteen Lemmas, Lemma 12.

If the first increment is defined cyclically, $\Delta q_0 = q_0\, q_{719}^{-1}$, then $\Delta q_{719} \cdots \Delta q_1 \Delta q_0 = 1$ for every sequence of states. A closure test must use the open product P.

Proof. The product telescopes to $q_{719}\, q_0^{-1}\, q_0\, q_{719}^{-1} = 1$. ■

A return to the identity at frame 720 therefore cannot purge drift: under the cyclic convention it happens whatever the drift. What removes drift is exact arithmetic, the integer clock of Section 2 and exact rational quaternions in the pump, so that P is computed without error and the memory test means something.

4 Euler's Formula in Hyperbolic Space

Status: **[PROVEN]**, with standard facts of hyperbolic geometry **[ESTABLISHED]** and one value **[COMPUTED]**. Sources: Traversal Monograph §§2, 5.1, 6.1; Fifteen Lemmas, Lemma 8 and Theorem 7.4; Hoard 2026e, Section 7.

On the circle Euler's formula wraps the line around a loop, so phases return. In hyperbolic space the same formula, with a square root of +1 in place of i, unrolls the line along a hyperbola, so nothing returns. Both formulas are one formula with a complex rapidity, and the Engine uses all three of its cases. This section states them, finds the golden rapidity of the {4,3,5} honeycomb, shows that the generator 1 + i is itself a hyperbolic screw, and derives the curvature and Euler characteristic of the Engine's hyperbolic cells.

4.1 The Hyperbolic Euler Formula

Let j satisfy $j^2 = +1$ with $j \neq \pm 1$, so that a + jb are the split-complex numbers, with the norm $a^2 - b^2$. The exponential series gives

$$e^{j\vartheta} = \cosh\vartheta + j\sinh\vartheta, \qquad \cos(i\vartheta) = \cosh\vartheta, \qquad \sin(i\vartheta) = i\sinh\vartheta.$$

Proposition 4.1 (The Hyperbolic Euler Formula). **[PROVEN]**

(1) As a matrix, $e^{\vartheta J}$ with $J = [[0, 1], [1, 0]]$ is $\cosh\vartheta\, I + \sinh\vartheta\, J$. It preserves the form $t^2 - x^2$ and is the Lorentz boost of rapidity ϑ; the point $(\cosh\vartheta, \sinh\vartheta)$ of the hyperbola lies at hyperbolic distance $|\vartheta|$ from (1, 0).

(2) The map $\vartheta \mapsto e^{j\vartheta}$ is injective: unlike $t \mapsto e^{it}$, it has no kernel.

(3) The units of norm ±1 form the group $\{\pm 1, \pm j\} \times e^{jR}$, isomorphic to the Klein four-group times R. Its elements of finite order are exactly ±1 and ±j; every other unit has infinite order.

Proof. (1) $J^2 = I$ splits the exponential series into its even and odd parts; $(\cosh\vartheta)^2 - (\sinh\vartheta)^2 = 1$. (2) cosh is even and sinh is injective. (3) A unit $a + jb$ with $a^2 - b^2 = 1$

and a > 0 is e^{jϑ} with ϑ = artanh(b/a); the other three branches of the two hyperbolas a² − b² = ±1 are its products with −1, j and −j. If (εe^{jϑ})^n = 1 with ε in the four-group, then e^{jnϑ} = ε^{−n} is ±1 or ±j with positive real part only when it is 1, so nϑ = 0. ■

The circle's unit group contains every root of unity; the hyperbola's contains four elements of finite order and no others. A hyperbolic phase never comes back: Proposition 2.5 had a kernel $2\pi\mathbb{Z}$ that let phases return, and part (2) says that here there is none. Section 5.1 turns this into the dividing line between what can and cannot be evaluated in constant time.

4.2 One Formula for Rotations and Boosts: Complex Rapidity

The Lorentz group $SO^+(3, 1)$ is double-covered by $SL(2, \mathbb{C})$, which acts on the boundary sphere of hyperbolic three-space by Möbius maps; $SU(2) \subset SL(2, \mathbb{C})$ is the stabilizer of a point of hyperbolic space, which is how the quaternions sit inside the Lorentz group (Fifteen Lemmas, Lemma 8, Proposition 6.3) **[ESTABLISHED]**. Along a fixed axis, rotations and boosts are one exponential:

$$\exp\left(\tfrac{\lambda}{2}\sigma_3\right) = \begin{pmatrix} e^{\lambda/2} & 0 \\ 0 & e^{-\lambda/2} \end{pmatrix}, \qquad \lambda = \vartheta + i\theta, \qquad \operatorname{tr} = 2\cosh\tfrac{\lambda}{2}.$$

Proposition 4.2 (Complex Rapidity). **[PROVEN]** for the formulas; the classification is **[ESTABLISHED]**.

(1) The matrix above acts on the boundary by z ↦ e^λ z and on hyperbolic space as a translation by ϑ along the axis from 0 to ∞ combined with a rotation by θ about it.

(2) It is elliptic when ϑ = 0 (a rotation), hyperbolic when θ = 0 (a boost) and loxodromic when both are nonzero (a screw).

(3) The spinor carries half of each: a rotation by θ has spinor entries e^{±iθ/2}, and a boost of rapidity ϑ has spinor entries e^{±ϑ/2}.

Proof. (1) The Möbius map of diag(e^{λ/2}, e^{−λ/2}) is z ↦ e^{λ/2}z/e^{−λ/2} = e^λ z. Its Poincaré extension (z, t) ↦ (e^λ z, e^ϑ t) to the upper half-space moves (0, 1) to (0, e^ϑ), a distance ϑ, and turns the vertical axis by θ. (2) and (3) read off the diagonal entries. ■

The imaginary part of λ is the angle of Section 2 and the real part is the rapidity of Section 4.1: Euler's formula on the circle and on the hyperbola are the two coordinate axes of one complex number. The Engine's spinor clock lives on the imaginary axis, its boosts and scale register on the real axis, and the generator 1 + i, as Section 4.4 shows, in between.

4.3 The Golden Rapidity of the {4,3,5} Honeycomb

A regular honeycomb {p, q, r} is hyperbolic when $\sin(\pi/p)\sin(\pi/r) < \cos(\pi/q)$. For cubes with five around each edge, {4,3,5}, this is $0.416 < 0.5$, and the honeycomb is compact (Coxeter 1973) **[ESTABLISHED]**. In the Klein model, where hyperbolic planes are flat, its central cell is a Euclidean-looking cube $[-a, a]^3$ (Fifteen Lemmas, Lemma 8, Proposition 6.4).

Proposition 4.3 (The Golden Cell). **[PROVEN]**; the volume is **[COMPUTED]** and agrees with the **[ESTABLISHED]** simplex volume.

Let φ = (1 + √5)/2 and a = φ^{−3/2} = 0.485868, so a² = √5 − 2.

(1) The cube [−a, a]³ of the Klein model has dihedral angles of 72°, so five cells meet at each edge.

(2) Its inradius is r_in = artanh a = 0.530638, and the boost B to the face-neighbour cell translates by ϑ = 2r_in, with

$$\cosh\vartheta = \varphi, \qquad \sinh\vartheta = \sqrt{\varphi}, \qquad e^{\vartheta} = \varphi + \sqrt{\varphi} = 2.890054, \qquad \tanh\frac{\vartheta}{2} = \varphi^{-3/2} = a.$$

(3) The spinor of B has half-rapidity ϑ/2 = r_in: the spin lift of a step to the next cell moves the spinor by exactly the distance from the centre of a cell to its face.

(4) √φ is a root of y⁴ − y² − 1, so the entries of B are algebraic integers of ℤ[√φ], and B can be applied in exact arithmetic.

(5) Each square face has angles α = arccos(1/√5) = 63.4349°, and the cell has volume 1.7224830 = 48 × 0.0358850633, forty-eight copies of the [4,3,5] Coxeter simplex.

Proof. Use the hyperboloid model with the form −tt′ + x·x′, in which the face x₁ = a of the Klein cube is the hyperplane x₁ = at with normal n = (a, 1, 0, 0). (1) For the faces x₁ = at and x₂ = at, ⟨n, n⟩ = ⟨m, m⟩ = 1 − a² and ⟨n, m⟩ = −a², so the interior angle has cosine a²/(1 − a²) = (√5 − 2)/(3 − √5) = (√5 − 1)/4 = cos 72°. (2) A point at Klein radius a on an axis is at distance artanh a from the centre, and the reflections in the plane x₁ = 0 and in the face compose to a translation by twice that distance; cosh(2 artanh a) = (1 + a²)/(1 − a²) = φ, so sinh ϑ = √(φ² − 1) = √φ, e^ϑ = φ + √φ, and tanh(ϑ/2) = sinh ϑ/(1 + cosh ϑ) = √φ/φ². (3) is Proposition 4.2(3). (4) (√φ)⁴ − (√φ)² − 1 = φ² − φ − 1 = 0. (5) The vertex figure of the cell is a spherical triangle with angles 72° and sides α, so cos α = cos 72°/(1 − cos 72°) = 1/√5; Appendix A, section M, recomputes α from tangent vectors and integrates the Klein volume element (1 − ‖x‖²)^{−2} over the cube. ■

The Traversal Monograph uses these boosts to carry the trefoil, scaled by 1/12, into the six face-neighbour cells. The scaled knot lies well inside the central cell: its largest coordinate is 0.25 and its largest Klein radius 0.2520, both below a = 0.4859 (Appendix A, section M).

4.4 The Eight-Step Clock as a Hyperbolic Screw

The Gaussian integer 1 + i acts in two ways. By conjugation, x ↦ qxq̄/N(q), its size cancels and only the rotation of Section 2.3 remains. By multiplication on the complex plane, the boundary of the upper half-space, its size survives.

Proposition 4.4 (The Generator as a Screw). [PROVEN]

(1) The map z ↦ (1 + i)z extends to the isometry (z, t) ↦ ((1 + i)z, √2 t) of the upper half-space, a loxodromic screw with complex rapidity

$$\lambda_{1+i} = \log(1+i) = \frac{\ln 2}{2} + i\,\frac{\pi}{4}.$$

(2) k steps translate by k(ln 2)/2 along the axis and twist by k·45°: four steps translate by ln 4 with a half-turn, matching (1 + i)⁴ = −4, and eight steps translate by ln 16 with no twist, matching (1 + i)⁸ = 16.

(3) The magnitude of (1 + i)^k carries exactly k/2 bits, so the bits removed by the gcd reduction of Proposition 2.2 equal the hyperbolic translation length divided by ln 2. Projective normalization is the quotient by the translation; what remains is the elliptic part, the spinor clock.

Proof. (1) The distance cosh d = 1 + (|z − z′|² + (t − t′)²)/(2tt′) is unchanged when z, z′ are multiplied by λ and t, t′ by |λ|; and λ = √2 e^{iπ/4}. (2) The distance from (0, 1) to (0, |λ|^k) is k ln|λ| = k(ln 2)/2, and the twist is k arg λ. (3) $\log_2|(1 + i)^k| = k/2 = (k(\ln 2)/2)/\ln 2$. ■

So the eight-step cycle of Section 3 is the elliptic shadow of a hyperbolic screw. The integer generator moves half a bit deeper with every step; the Engine keeps only the angle by dividing out the depth, and that division is what keeps the clock in two signed bits.

4.5 The Scale Register as a Boost

Proposition 4.5 (Dilation Is a Boost). [PROVEN]

In R^{4,1} the map D_λ = diag(1, 1, 1, λ, 1/λ) preserves the form of Section 3.2, and λ D_λ X(x) = X(λx). A dilation of three-space by λ is therefore a boost of rapidity ln λ in the plane of e_∞ and e_0, and the dyadic scale level k of the Einstein Tile's scale register (2^k C_0) is the rapidity k ln 2.

Proof. The form pairs the last two coordinates as −αβ′ − βα′, which is unchanged when α is multiplied by λ and β by 1/λ. And λ(x, λ‖x‖²/2, 1/λ) = (λx, ‖λx‖²/2, 1). In the basis e_0 ± e_∞/2 this is cosh(ln λ) I + sinh(ln λ) J. ■

In the conformal model, rotations of three-space are elliptic, dilations are hyperbolic and translations are parabolic. Two steps of the clock, $(1 + i)^2 = 2i$, are one dyadic level together with a quarter turn of spin: the clock and the scale register advance the imaginary and real parts of the same complex rapidity.

4.6 Euler's Polyhedron Formula and Gauss–Bonnet

Euler's name is on a third formula, V − E + F = 2 for a convex polyhedron, and in hyperbolic space it becomes a statement about curvature.

Proposition 4.6 (Curvature of the Cell). [PROVEN], using the Gauss–Bonnet theorem **[ESTABLISHED]**.

(1) A hyperbolic polygon with n vertices and angles α_1, …, α_n has area (n − 2)π − Σα_i. Each square face of the {4,3,5} cell has area 2π − 4α = 1.854590.

(2) The boundary of the cell has total curvature 4π = 2πχ(S²): the six faces contribute −6(2π − 4α) and the eight vertices the angle defects 8(2π − 3α). The identity holds for every α and is Euler's formula 8 − 12 + 6 = 2.

Proof. (1) is Gauss–Bonnet for a geodesic polygon of curvature −1. (2) −6(2π − 4α) + 8(2π − 3α) = 4π + 24α − 24α = 4π; in general the sum of face terms and vertex defects is 2π(V − E + F). ■

The Euclidean cube puts all its curvature at the eight corners, π/2 each; the hyperbolic cube moves part of it into the faces, which is why its face angles are 63.43° and not 90°.

Proposition 4.7 (Euler Characteristic of a Closed {4,3,5} Manifold). **[PROVEN]**; existence **[ESTABLISHED]**.

If a closed three-manifold is tiled by C cubes of the {4,3,5} honeycomb, it has V = 2C/5 vertices, E = 12C/5 edges and F = 3C faces, so χ = V − E + F − C = 0, and C is a multiple of 5. Such manifolds exist.

Proof. Each cube has 8 vertices, 12 edges and 6 faces. Twenty cubes meet at each vertex, since the vertex figure is an icosahedron with 20 faces; five meet at each edge and two at each face. So V = 8C/20, E = 12C/5, F = 6C/2, and 2/5 − 12/5 + 3 − 1 = 0; V and E are integers only if 5 | C. The symmetry group [4,3,5] is a cocompact discrete group, and by Selberg's lemma it has torsion-free subgroups of finite index, whose quotients are closed manifolds tiled by the cells. ■

The vanishing agrees with the general fact that every closed odd-dimensional manifold has Euler characteristic 0 **[ESTABLISHED]**. A finite quotient of the honeycomb is the hyperbolic analogue of the Engine's cubic torus $(\mathbb{Z}/4)^3$, and the Ramanujan graphs of Section 6.3 are the analogous finite quotients of the norm-5 tree.

4.7 Exponential Volume

Proposition 4.8 (Volume and Address Growth). **[PROVEN]**, with part (1) **[ESTABLISHED]**.

(1) In hyperbolic three-space a ball of radius R has volume π(sinh 2R − 2R) and a sphere has area 4π sinh²R, so V(R + 1)/V(R) → e² = 7.389.

(2) The norm-5 quaternions 1 ± 2i, 1 ± 2j, 1 ± 2k generate a free group; its reduced words of length L number 6 · 5^{L−1}, and their products are distinct quaternions of norm 5^L (Hoard 2026e, Theorem 7.2).

(3) Under the abelianization 1 ± 2u_k ↦ ±e_k, the words of length L land exactly on the points v of $\mathbb{Z}^3$ with $\|v\|_1 \le L$ and $\|v\|_1 \equiv L \pmod 2$, except the origin when L = 2. Their number, the cubic shadow, is

$$N_{\text{shadow}}(L) = \begin{cases} \dfrac{2L^3 + 6L^2 + 7L + 3}{3}, & L \text{ even},\ L \neq 2, \\ \dfrac{(L+1)\left(2(L+1)^2 + 1\right)}{3}, & L \text{ odd}, \end{cases} \qquad N_{\text{shadow}}(2) = 18,$$

about (2/3)L³, half of the (2L + 1)(2L² + 2L + 3)/3 points of the full ℓ^1 ball.

Proof. (1) Integrate the area 4π sinh²r. (3) Each letter changes $\|v\|_1$ by ±1, which gives the bound and the parity. Conversely, let v ≠ 0 have $\|v\|_1 = L - 2t$ and let w be a monotone word for v, which is reduced. If some coordinate of v is 0, say v_b, then e_b^t w e_b^{−t} is reduced of length L. If none is, order w as its e_2-steps, then its e_1-steps, then its e_3-steps, with $v_1 > 0$ after a sign change, and use $e_1^t w e_1^{-t}$. For v = 0 and L = 2t with t ≥ 2, take $e_1^{t-1} e_2 e_1^{-(t-1)} e_2^{-1}$. A word of length 2 with image 0 would be g g^{−1}, which is not reduced. Summing the sphere sizes 4k² + 2 over $k \equiv L \pmod 2$ gives the closed forms. ■

Appendix A, section R, confirms part (3) by enumeration for L ≤ 8 and by dynamic programming for L ≤ 20. The tree fills its addresses at the rate 5 per step, the hyperbolic ball

at e^2 per unit of radius, and the cubic shadow only polynomially; Section 6.1 tabulates the ratio, and Section 5.7 explains why this growth is a cost for search, not a resource.

4.8 The Clock and the Depth

Proposition 4.9 (The Engine's Screw Motion). **[PROVEN]** Fifteen Lemmas, Lemma 5 Part 2 and Lemma 8.

(1) The rotations of the core cube, the group O, are exactly the rotations of the Klein model that preserve the central cell of the {4,3,5} honeycomb, and their spin lift 2O contains the eight-step clock: the clock is a symmetry of the central hyperbolic cell.

(2) The face-neighbour boost B along the x-axis commutes with Rot_x(90°), and their composite is a screw with complex rapidity $\vartheta + i\pi/2$: one step of the clock and one cell of depth.

(3) Repeating the screw, the angle returns after four steps (and the spinor after eight), while the depth grows by $\vartheta = 1.0613$ per step without return.

Proof. (1) A rotation of R^3 fixing the origin acts on the Klein model as a hyperbolic isometry, and it preserves $[-a, a]^3$ exactly when it preserves $[-1, 1]^3$; Proposition 2.4(3) places the clock in 2O. (2) The boost acts in the plane of t and x_1, the rotation in the plane of x_2 and x_3, so they commute, and their complex rapidities add. (3) Proposition 4.1(2) and Proposition 2.3. ■

The Engine's motions thus fall into Euler's three cases. Elliptic motions close and need only a residue to store. Hyperbolic motions never close, fill exponential volume and need ever more bits. Loxodromic motions do both at once. Section 5 shows that the conditional collapse of P and NP lives entirely in the first case.

5 Conditional P = NP: What the Cycle Makes Constant and What It Cannot

Status: **[PROVEN]** for every theorem and proposition; the machine model and classes are **[DEFINITION]**; P versus NP is **[OPEN]**. Sources: Traversal Monograph §§2–5 and 7; Hyperbolic Horizon, Theorem 3; Fifteen Lemmas, Lemmas 7, 9, 12, 13, 14 and 15; Hoard 2026e.

The Traversal Monograph states that under five axioms the classes P_GET, NP_GET and TIME_GET(O(1)) coincide, and tags the statement **[PROVEN]**. This section separates what is proven in that statement from what is not. The proven core is a theorem about structure: search becomes evaluation when the answer is a procedural function of the question. Its constant-time form holds for finite-order motions and inside bounded horizons, where it holds for every problem. Its classical form is an exact equivalence whose condition is the P versus NP problem itself.

5.1 The Euler Evaluation Dichotomy

The question underneath every constant-time claim of the Engine is when an N-step evolution g^N can be evaluated without taking N steps. Euler's formula answers it.

Theorem 5.1 (The Euler Evaluation Dichotomy). **[PROVEN]** Fifteen Lemmas, Lemma 13 Parts 1 to 3; Propositions 2.4 and 4.1 here.

Let g act on states and be represented exactly, with integer or rational coordinates in a fixed basis.

(1) If g has finite order n, then g^N = g^{N mod n}. With the n powers of g in a table, g^N costs one reduction modulo n and one lookup, a constant number of word operations for every N < 2^64. The eight-step clock (n = 8, projectively), the tier spinor h (n = 6) and the 24 Hurwitz units are of this kind.

(2) If g is a rational unit quaternion of infinite order, the reduced denominator of g^N is unbounded; for g = (3 + 4i)/5 it is exactly 5^N. Then g^N needs at least N $\log_2 5$ bits, and no algorithm outputs it exactly in a constant number of word operations for all N. Repeated squaring computes it with O(log N) multiplications of Θ(N)-bit numbers.

(3) If g is hyperbolic, with an eigenvalue $e^{\vartheta} > 1$ and algebraic-integer entries, the entries of g^N grow like e^{Nϑ}, and in any fixed integral basis they need $N\vartheta/\ln 2 - O(1)$ bits: 2.543 bits per step for the integer boost [[3, 4], [2, 3]] of the form $t^2 - 2x^2$, and 1.531 bits per step for the golden face boost of Proposition 4.3.

(4) In floating point, Euler's formula g^N = cos Nθ + u sin Nθ costs a constant number of operations for every N, but the error grows with Nθ.

Exact evaluation of an N-step homogeneous evolution in constant time is therefore possible exactly when the generator has finite order, or when N is bounded.

Proof. (1) is division with remainder. (2) $3 + 4i = (2 + i)^2$ and $5 = (2 + i)(2 - i)$ in the Gaussian integers, which factor uniquely; (2 + i)^{2N} is divisible by no power of 2 − i, hence not by 5, so (3 + 4i)^N/5^N is in lowest terms. An output with denominator 5^N has at least N $\log_2 5$ bits, and a constant number of 64-bit words holds a bounded number of bits. (3) The spectral radius of g^N is e^{Nϑ} and is at most a constant times its largest entry; an algebraic integer of absolute value X has a coordinate of size at least X/C in a fixed integral basis. The two rates are $\log_2(3 + 2\sqrt{2})$ and $\log_2(\varphi + \sqrt{\varphi})$. (4) The argument Nθ is held with an absolute error of order Nθ · 2^{−53}; Appendix A, section Q, measures errors of 6.2 × 10^{−14} at $N = 10^3$ rising to 5.0 × 10^{−5} at $N = 10^{12}$. ■

Corollary 5.2 (The Three Sources of Constant Time). **[PROVEN]**

Every constant-time operation of the Engine is of one of three kinds: a finite-order residue (Theorem 5.1(1)); a telescoping identity that returns the two endpoints of a chain and nothing else (Proposition 3.9); or exact arithmetic inside a bounded precision horizon (axiom C5). None of the three performs a search.

5.2 The Machine Model and the Classes

Definition 5.3 (The Geometric Engine Machine). **[DEFINITION]** Traversal Monograph, Definition 2.1; Fifteen Lemmas, Lemmas 4, 5, 8 and 10.

The machine M_GET has six components:

1. *the lower manifold, the voxel cube $V = [-2C_0, 2C_0]^3$ around the core $K = [-C_0, C_0]^3$, with rotation group O and spin lift 2O;*
2. *the upper manifold, five toroidal tiers with $R_n = (5n^2 + 15n + 15)C_0$, $r_n = (2n + 4)C_0$ and windings $W_n = 16 - 2n$ summing to $W_\Sigma = 60$, in disjoint shells separated from the lower manifold by a gap of $(11 - 2\sqrt{3})C_0 = 7.536C_0$;*
3. *the 64-bit Einstein Tile: a core word (bits 0 to 31) in one of sixteen forms, a scale register and control nibbles (bits 32 to 47), and a gate byte and South Halo header (bits 48 to 63) that include the three-bit cycle step;*
4. *the twistor operator: the conformal lift X of Section 3.2, the rational ghost map $G(p) = ((1 + \|p\|^2), 2p)/(1 - \|p\|^2)$ into the hyperboloid, and the {4,3,5} face-neighbour boosts of Section 4.3;*
5. *the $720 \to 72 \to 10$ pump of Section 3.6;*
6. *the Dual Helix: Helix A proposes candidates by geometry or diffusion, and Helix B certifies them by exact arithmetic.*

Cost model: a word RAM with 64-bit words, in which each arithmetic, logical or memory operation on a word, and each lookup in a stored table, costs one unit. The size and the precomputation of tables are counted separately.

The Traversal Monograph does not state a cost model; the last clause supplies one, because without it "O(1)" has no meaning.

Definition 5.4 (The Classes). **[DEFINITION]** Traversal Monograph, Definition 3.1.

SEQ(N) is the class of evolutions $E(N) = T_N \cdots T_1 E(0)$ evaluated by N serial operations, and of searches of a tree of depth L by exhaustion. NP_GET is the class of state-reachability and constraint problems on M_GET whose candidate witness, a tile or a tuple of tiles, Helix B verifies in O(1) word operations. P_GET = TIME_GET(O(1)) is the class whose terminal state or witness is synthesized in O(1) word operations, independent of N.

5.3 The Five Integrability Axioms

Definition 5.5 (Geometric-Procedural Integrability). **[DEFINITION]** Traversal Monograph, Definition 3.2.

A problem satisfies C_GET if it satisfies:

- *C1, procedural homomorphism: either (C1a) a shared seed P gives an O(1) map $N \mapsto (x_N, y_N, z_N)$ and tables L_1, L_2, L_3 with $E(N)$ assembled from $L_1[x_N]$, $L_2[y_N]$, $L_3[z_N]$; or (C1b) the increments are coboundaries $\Delta q_n = q_n$ q_{n−1}^{−1}, or pass through the pump of Section 3.6;*
- *C2, linearization: metric constraints are carried into the null cone (Proposition 3.3) or into the {4,3,5} honeycomb by the ghost map;*
- *C3, spinor and parity conservation: cross-tier transfers use the tier spinor $h = (1 - i + j + k)/2$ of order 6, with $h\, h\, \bar{h}\, \bar{h} = 1$ and the tier rotations closing to the identity, and at most one record of a parity group $H_1H_2H_3H_4 = 1$ is lost;*

- *C4, moment locking: the five-moment Vandermonde system across the gap is invertible and its nonlinear coupling lies in a contraction ball (Fifteen Lemmas, Lemma 6);*
- *C5, bounded horizons: the states stay within the exact fixed-width horizons of the tile: Form 1 norms N(Q) ≤ 16,129; quadrances up to Q_phys = 2.29 × 10^7 for cubit-anchored lattices; the Form 2 range M = 881 × 883 × 887 = 690,017,701; and tree depth L ≤ 54.*

5.4 The Conditional Collapse

Theorem 5.6 (Conditional Collapse). **[PROVEN]** as stated here; Traversal Monograph, Theorem 3.3, restated with its hypotheses made explicit.

Let a problem satisfy C1 to C5, and let the tables of C1a be given. Then, in the cost model of Definition 5.3:

(1) the state E(N) is assembled in O(1) word operations and three lookups;

(2) the pump holonomy H = P^{−1} = q_0 q_{719}^{−1} is evaluated from two stored states, and a chain of k independently generated increments in O(log k) parallel depth with O(k) work;

(3) one erased record of a parity group, or one erased branch of the recombination of Section 3.7, is recovered in O(1) group operations;

(4) a Form 2 state X < M is recovered from its residues modulo 881, 883 and 887 by Garner's algorithm with closed-form inverses in O(1) arithmetic;

(5) tier offloads by h preserve norms exactly, N(hq) = N(q), with no bit growth.

Hence every problem of NP_GET that satisfies C_GET lies in P_GET.

Proof. (1) is the definition of C1a followed by two quaternion products. (2) is Proposition 3.9, and for independent increments a balanced prefix product of depth [$\log_2$ k] (Fifteen Lemmas, Lemma 12 Part 3). (3) is Proposition 3.10. (4) is Fifteen Lemmas, Lemma 9 Part 2. (5) N(h) = 1 and the norm is multiplicative. ■

The theorem is true, and every step of it is a finite-order residue, a telescoping identity or bounded arithmetic, as Corollary 5.2 says. What it does not say is decided by its hypothesis. Hyperbolic Horizon, Theorem 3, asserts that any O(N) trajectory governed by shared procedural parameters decomposes as C1a requires. By Theorem 5.1 that holds when the generators have finite order or when the tables already hold the needed powers, and fails for infinite-order or hyperbolic generators over unbounded N **[SUPERSEDED]** as a general statement (Section 9.3).

5.5 What the Collapse Is

Proposition 5.7 (Search Replaced by Evaluation). **[PROVEN]**

Under C1a the witness is W = Φ(L_1[x_N], L_2[y_N], L_3[z_N]) for a fixed O(1) map Φ. Finding W and verifying a candidate both reduce to evaluating this expression, so P_GET = NP_GET on the problems satisfying C_GET holds by construction. Its content

lies entirely in the hypothesis that a procedural map and its tables exist for the problem at hand.

Proof. Evaluate the expression; to verify a candidate, evaluate and compare. Neither step uses anything beyond C1a. ■

Proposition 5.8 (Finite-Horizon Triviality). [PROVEN]

Under C5 the instances of any problem form a finite set S. Every function on a finite set is computed in O(1) by a table with |S| entries. Constant time under C5 is therefore not a complexity statement: complexity classes describe how cost grows with input size, and C5 bounds the size.

Proof. Store f(s) at address s. ■

The cost moves into the table: a single 64-bit tile has 2^{64} values, and a table indexed by it needs 2^{64} entries. The Engine's tables are small only because its structures are periodic, which is Theorem 5.1(1) again.

Proposition 5.9 (Constant Time Is Not P). [PROVEN]

On inputs of unbounded length n, an algorithm that runs in O(1) word operations reads O(1) words, so its answer depends on at most a constant number of input bits. The parity of n bits is in P and not in TIME(O(1)); so is satisfiability: changing one unread clause of $x_1 \wedge x_2 \wedge \ldots \wedge x_m$ into $\neg x_1$ changes the answer. Hence P_GET = TIME_GET(O(1)) is a proper subclass of P, and NP-complete problems on unbounded inputs are not in it, whether or not P = NP.

Proof. An algorithm that reads c words cannot distinguish two inputs that agree on those words; for each of the two problems, inputs of any length $n > cw$ exist that agree on any given cw positions and have different answers. ■

5.6 The Conditional Equivalence

The classical question survives in a precise form. Replace constant time by polynomial time, and the seed P of C1a by a seed computed from the instance.

Definition 5.10 (Integrable Encoding). [DEFINITION]

Let L be a language in NP with polynomial-time verifier V. An integrable encoding of L is a pair of polynomial-time maps, enc taking an instance x to procedural parameters P_x of polynomial size, and eval taking P_x to a candidate witness w, such that V(x, eval(enc(x))) = 1 for every x in L.

The pair (enc, eval) is the polynomial analogue of C1a: enc plays the shared seed, eval the lookups and their assembly.

Theorem 5.11 (The Conditional P = NP Equivalence). [PROVEN]

The following are equivalent:

(a) P = NP;

(b) every language in NP has an integrable encoding;

(c) satisfiability (SAT) has an integrable encoding.

Proof. (a) ⇒ (b): if P = NP, the language of pairs (x, u) such that some witness of x extends the prefix u is in NP, hence in P, and fixing the witness one bit at a time finds a witness of any x in L with polynomially many calls. Take enc(x) = x and eval this procedure. (b) ⇒ (c) is immediate. (c) ⇒ (a): to decide whether a formula x is satisfiable, compute w = eval(enc(x)) and accept exactly when V(x, w) = 1. This is correct, since an unsatisfiable formula has no witness, and takes polynomial time; SAT is NP-complete (Cook 1971; Levin 1973), so P = NP. ■

This is the exact sense in which the Engine yields a conditional P = NP equivalent. The Engine's axioms describe integrable structure, and the conditional collapse holds wherever that structure is given. Demanding it of every problem in NP, with the encoding computed from the instance, is exactly as strong as P = NP. Whether satisfiability has an integrable encoding is the P versus NP problem **[OPEN]**. Nothing in the Engine supplies one, and the theorem neither advances nor blocks a resolution; it says what a proof of P = NP by the Engine's methods would have to construct. Appendix A, section T, runs the step (a) ⇒ (b) on random satisfiable 3-SAT formulas with 14 variables: a witness is found with n + 1 = 15 calls to a decision procedure.

5.7 Verification and Search in the Hyperbolic Tree

The norm-5 tree makes the asymmetry of NP geometric.

Proposition 5.12 (Verifying an Address). [PROVEN] Hoard 2026e, Theorem 7.2.

An address q of norm 5^L determines its reduced word, which is recovered by L right-divisions, each identifying the unique generator g with q ḡ divisible by 5. Verification of an address costs O(L) operations.

Proposition 5.13 (Searching for an Address). [PROVEN]; the quantum bound is **[ESTABLISHED]**.

If the target at depth L is specified only by a membership test, every deterministic algorithm needs all 6 · 5^{L−1} tests in the worst case, every randomized algorithm needs about half of them on average for a uniformly placed target, and every quantum algorithm needs Ω(√(6 · 5^{L−1})) (Bennett et al. 1997).

Proof. An adversary answers no to every test until one address is left untested, and places the target there. For a uniform target, a randomized order finds it after (N + 1)/2 tests on average. ■

Verification costs the depth, search costs the volume. Exponential volume (Proposition 4.8) is a cost for search, not a resource: the tree offers 6.66×10^{37} addresses at depth 54, and finding one by its property alone takes as many tests. The Engine closes the gap only when the target's address is a procedural function of the instance, which is C1a again.

5.8 The Boundary: Seven Obstructions

Theorem 5.14 (Where Constant-Time Collapse Stops). [PROVEN]

(I) Pigeonhole. A map from a set of size S into 64-bit words has a fibre of size at least $S/2^{64}$. Twelve nested layers of the 64-voxel cube have $64^{12} = 2^{72}$ states, so a 64-bit summary has fibres of at least 256; an 8-bit header computed from a 56-bit payload

has buckets of 2^{48} on average. A header filters; a Merkle root certifies (Fifteen Lemmas, Lemma 9 Part 4(a) and Lemma 15).

(II) History loss. The 13,824 three-step chains of Hurwitz units give 24 products; two lost records of a parity group leave a one-parameter family; telescoping keeps endpoints only (Proposition 3.10). The norm-5 tree keeps history, but at $\lceil \log_2(6 \cdot 5^{L-1}) \rceil = \Theta(L)$ bits, 126 bits at $L = 54$.

(III) Niven and bit growth. Rational finite orders are 1, 2, 3, 4 and 6; a product of k integer quaternions with coordinates below 2^b has coordinates below $2^{k(b+2)-2}$ (Fifteen Lemmas, Lemma 13; Theorem 5.1(2)).

(IV) Torsion-free triviality. Every homomorphism from a finite group into a torsion-free group is zero, so a cyclic register can be stored only as a residue, never as a linear invariant (Proposition 2.5; Fifteen Lemmas, Lemma 9 Part 4(b)).

(V) The Circularity Boundary. A drift without a potential carries a current but leaves the stationary state uniform; selecting a target set requires evaluating the target indicator (Fifteen Lemmas, Lemma 14 Part 4; Hoard 2026e, Propositions 2.2 to 2.6).

(VI) Hyperbolic non-return. No hyperbolic motion has finite order, and exact boosts grow by a fixed number of bits per step (Proposition 4.1; Theorem 5.1(3)).

(VII) Search and size. Black-box search in the tree costs its volume (Proposition 5.13), and constant time on unbounded inputs reads a bounded part of the input (Propositions 5.8 and 5.9).

Proof. Each part is proved where cited. ■

Each obstruction is a theorem about its own object. None is a lower bound for satisfiability, and none decides P versus NP; together they mark where the Engine's constant-time methods stop.

5.9 Barriers

Baker, Gill and Solovay (1975) constructed oracles A and B with $P^A = NP^A$ and $P^B \neq NP^B$ **[ESTABLISHED]**.

Proposition 5.15 (Lookup Arguments Relativize). **[PROVEN]**, using the result just cited.

The proof of Theorem 5.6 remains valid when the tables L_1, L_2, L_3 are replaced by arbitrary oracles. An argument built only from lookups and the operations of Definition 5.3 therefore relativizes, and by Baker, Gill and Solovay no relativizing argument settles P versus NP.

Proof. Each step of Theorem 5.6 uses the tables only through queries, so it holds verbatim relative to any oracle. A relativizing proof of $P = NP$ would give $P^B = NP^B$, and one of $P \neq NP$ would give $P^A \neq NP^A$. ■

Two further barriers, natural proofs (Razborov and Rudich 1997) and algebrization (Aaronson and Wigderson 2009), constrain arguments for $P \neq NP$ **[ESTABLISHED]**. A resolution by the Engine would have to escape all three.

5.10 The Constant-Time Pipeline

Table 6 gives the Traversal Monograph's instruction budget for synthesizing and certifying one state inside C_GET. It is a design estimate: no implementation has been timed.

Table 6: The pipeline of one constant-time synthesis.

Stage	Operation	Cost	Status
1. South Halo pre-filter	Extract bits 56 to 63 and compare	1 shift, 1 compare	**[HEURISTIC / COMPUTED]**
2. Form and coordinate map	Read the form ID; evaluate N ↦ (x_N, y_N, z_N) or Garner's CRT	5 to 10 operations	**[PROVEN]** for Form 2
3. Three lookups	Q_i = L_i[·]	3 memory reads	**[PROVEN]** under C1a
4. Coherence and assembly	AND and XOR on the gate byte; two quaternion products; apply the scale register	32 multiplications, 24 additions, 2 logical operations	**[COMPUTED]**
5. Holonomy or parity	H = $q_0\, q_{719}^{-1}$ or $H_4 = (H_1H_2H_3)^{-1}$	1 to 3 quaternion products	**[PROVEN]**
6. Helix B certificate	Norm; Form 1 bitmap read; Gram form of the TQF	about 10 operations	**[PROVEN]** for Form 1
Total		at most about 60 operations on the Form 1 path	**[HEURISTIC / COMPUTED]**

The total of about 60 holds on the Form 1 path, where the certificate is one read of the 2,017-byte prime bitmap. On the cubit and wide horizons Helix B uses a deterministic Miller–Rabin test with 12 bases, each a modular exponentiation of about 64 squarings, so the certificate costs hundreds of operations: still constant, since the horizon is fixed, but not 60.

5.11 Where the Equivalence Holds

Table 7: Regimes of the Engine, their cost and their status.

Regime	Mechanism	Time	Space	Status
Procedural evolution with given tables	C1a: three lookups and assembly	O(1)	three tiles; tables of size M	**[PROVEN]** under C1a
720-frame coboundary chain	Telescoping, H = $q_0\, q_{719}^{-1}$	O(1)	the stored endpoints	**[PROVEN]**
Chain of k independent increments	Balanced prefix product	O(log k) depth, O(k) work	O(k(b + 2)) bits	**[PROVEN]**
Cross-tier offload	Norm-preserving spinor h	O(1) per tier	no growth	**[PROVEN]**
One erased record or branch	Group inverse or pulled-back sum	O(1)	one tile	**[PROVEN]**
Cubic addressing below M	Closed-form Garner CRT	O(1)	30 bits	**[PROVEN]**
Tree address at depth L	Shift-and-add synthesis; right-division	O(L)	Θ(L) bits	**[PROVEN]**

Regime	Mechanism	Time	Space	Status
Diffusive mixing on n nodes	Ramanujan quotient X(5, q)	O(log n)	O(log n) bits per node	**[ESTABLISHED]**, **[COMPUTED]**
Unstructured search at depth L	Membership tests only	6 · 5^{L−1} tests	—	**[PROVEN]**
SAT and all of NP	Integrable encoding (Theorem 5.11)	polynomial if and only if P = NP	—	**[OPEN]**

6 Multi-Scale Predictions

Status: **[COMPUTED]**, with extrapolations and horizons **[HEURISTIC / COMPUTED]**. Sources: Traversal Monograph §6; Hoard 2026e, Sections 7 and 8; Fifteen Lemmas, Lemma 11.

The Traversal Monograph closes with five tables of predictions. This section recomputes four of them from the results above; where a recomputed value differs from the monograph's, Section 9.3 lists the difference. The fifth, on multi-engine memory, rests on the Fifteen Lemmas, §§7.4–7.7, and its only law used here is the single-erasure law of Proposition 3.10.

6.1 Hyperbolic Addresses Against the Cubic Shadow

Table 8 compares the 6 · 5^{L−1} addresses at depth L of the norm-5 tree with the N_shadow(L) points of $\mathbb{Z}^3$ that its words reach (Proposition 4.8). Coordinates of an address of norm 5^L are at most 5^{L/2} in absolute value.

Table 8: Tree addresses, cubic shadow and word widths by depth.

Depth L	N_tree = 6 · 5^{L−1}	N_shadow(L)	Ratio	Norm 5^L	Largest coordinate	Signed bits	Regime
1	6	6	1.00	5	2.24	3	—
2	30	18	1.67	25	5	4	30 addresses, as many as the nodes of the 30-wheel **[CORRESPONDENCE]**
5	3,750	146	25.7	3,125	55.9	7	Form 1 doubled core
6	18,750	231	81.2	15,625	125	8	Signed-byte coordinates; norm below 16,129
8	468,750	489	958.6	390,625	625	11	—
10	1.172×10^{7}	891	1.315×10^{4}	9.766×10^{6}	3,125	13	Cubit horizon, Q ≤ 2.29×10^{7}
12	2.930×10^{8}	1,469	1.994×10^{5}	2.441×10^{8}	15,625	15	Form 2 range, M = 6.90×10^{8}
17	9.155×10^{11}	3,894	2.351×10^{8}	7.629×10^{11}	8.735×10^{5}	21	Source-note horizon, Q ≈ 10^{12}
27	8.941×10^{18}	14,644	6.105×10^{14}	7.451×10^{18}	2.729×10^{9}	33	Norm below 2^{63}
54	6.661×10^{37}	110,935	6.005×10^{32}	5.551×10^{37}	7.451×10^{18}	64	Coordinates below 2^{63}

Prediction 1. **[COMPUTED]** At depth 54 the tree holds $6.661 \times 10^{37} \approx 2^{125.6}$ distinct addresses, each held exactly in four 64-bit lanes, while the words of that length reach 110,935 points of the cubic lattice and walks of at most 54 steps reach 215,929. The ratio 6.0×10^{32} compares two counts. It is not a compression: naming one address still takes 126 bits, and finding one by a property takes the search of Proposition 5.13.

6.2 Precision Horizons

Table 9: Horizons of exact arithmetic in the Einstein Tile (Hoard 2026e, Section 8.2; Traversal Monograph, §6.2).

Tier	Largest quadrance	Coordinates	Gram TQF width	Register	Certification
1. Form 1 core	16,129 (norm); 258,064 (differences)	127 in doubled units	36 bits	int64	2,017-byte prime bitmap, one read
2. Cubit horizon	2.29×10^7 = u_rel^{−2}	4,785 C_0, about 2.57 km	53 bits	int64	Deterministic Miller–Rabin, 12 bases
3. Form 2 CRT range	$X < 6.90 \times 10^8$	30-bit payload as three 10-bit residues	addressing only	uint64	Closed-form Garner, then Miller–Rabin
4. Source-note horizon	10^{12}	10^6	about 80 bits (85 for the S^2 form)	128-bit	Deterministic Miller–Rabin, 12 bases

The widths are **[COMPUTED]** and the horizons that set them **[HEURISTIC / COMPUTED]**, since they rest on the resolution rule $\delta\theta \approx 1/\sqrt{Q}$ (Hoard 2026h, Problem 10).

Prediction 2. **[PROVEN]** The 30-node wheel rejects 22 of every 30 integers in O(1) before certification, 73.3 %, with a proposer bias of 30/8 = 3.75, the largest of any primorial wheel whose unit classes fit a 3-bit field; on Form 1 Hurwitz cores it rejects 65.74 % (Hoard 2026e, Propositions 3.2 and 3.3).

6.3 Mixing on Ramanujan Graphs and Cubic Tori

The norm-5 tree reduced modulo a prime $q \equiv 1 \pmod 4$ is the 6-regular Ramanujan graph X(5, q) of Lubotzky, Phillips and Sarnak, on PSL(2, $\mathbb{F}_q$) when 5 is a square modulo q and on PGL(2, $\mathbb{F}_q$) otherwise **[ESTABLISHED]**. Its nontrivial eigenvalues are at most $2\sqrt{5}$, so its spectral gap is at least $6 - 2\sqrt{5} = 1.528$, while the 6-regular cubic torus of side m has gap $2(1 - \cos(2\pi/m))$, about $4\pi^2/m^2$. Table 10 gives the time for the continuous-time walk with generator $A - 6I$ to come within 1 % of uniform in total variation, on a grid of 0.05.

Table 10: Mixing times to 1 % on Ramanujan graphs and cubic tori of nearly equal size.

q	Group	n	t_mix on X(5, q)	Torus	n	t_mix on torus	Ratio
13	PGL	2,184	3.25	13^3	2,197	20.15	6.2
29	PSL	12,180	3.80	23^3	12,167	62.15	16.4
61	PSL	113,460	4.20	48^3	110,592	269.30	64.1
101	PSL	515,100	4.55	80^3	512,000	747.35	164.3
1009	PSL	513,621,360	about 6.2	800^3	5.12×10^8	about 74,700	about 1.2×10^4

The first four rows are **[COMPUTED]** (Appendix A, section U); the first two reproduce Hoard (2026e), Table 10. The last row is **[HEURISTIC / COMPUTED]**: the torus time scales as m^2 ($269.30 \times (80/48)^2 = 748.1$ against the computed 747.35), and the Ramanujan time grows by about 0.24 per unit of ln n across the computed rows.

Prediction 3. **[COMPUTED]** and **[HEURISTIC / COMPUTED]** On half a million nodes the Ramanujan quotient mixes 164 times faster than the cubic torus, and the factor grows like n^{2/3}/ln n, to about 10^4 at half a billion nodes. This is a speedup of a randomized proposer; it changes no problem's complexity class.

6.4 Clock Windows of the Cycle

Table 11: Pulse counts of 1/60 s and the clocks that close on them.

Pulses	Time	Caesium-133 periods	40 Hz periods	432 Hz periods	What closes
1	1/60 s	153,210,529.5	—	—	Nothing; the half-period is the parity residual
2	1/30 s	306,421,059	—	—	Caesium
6	0.1 s	919,263,177	4	—	Caesium and 40 Hz
10	1/6 s	1,532,105,295	—	72	Caesium and 432 Hz
24	0.4 s	3,677,052,708	16	172.8	The joint 24-frame register, caesium and 40 Hz
30	0.5 s	4,596,315,885	20	216	All three clocks
60	1 s	9,192,631,770	40	432	All three; $W_\Sigma = 60$ pulses
120	2 s	18,385,263,540	80	864	The joint register and all three clocks (Proposition 3.8)
720	12 s	110,311,581,240	480	5,184	The outer gear: 6 such closures, 30 joint cycles, 90 spinor cycles

All entries are **[COMPUTED]** under the frame convention of Proposition 3.8. The Traversal Monograph's further window of 25,920 pulses (432 s), called a Platonic Great Year, is the number of years in the astronomical precession cycle of that name; the match is **[CORRESPONDENCE]**.

7 Relation to the Geometric Engine of Time

Status: as tagged in each row. Sources: Hoard 2026a; Fifteen Lemmas; Sections 2 to 6 here.

The zero-point cycle joins three structures of the Engine that earlier installments treated apart: the six channels of the orbital calculator, the spinor clock of the Einstein Tile, and the hyperbolic geometry of the twistor operator. One exact link is new.

> **Proposition 7.1 (The Six Handshakes Are the Six Channels).** **[PROVEN]**; the identification with the calculator's channels is **[CORRESPONDENCE]**.
>
> *The six handshake events of a joint 24-frame cycle fall at the frames $n = 4m + 2$, $m \in \mathbb{Z}/6$. The map $m \mapsto (j, \sigma)$, where $j = n \bmod 3$ is the branch and $\sigma = (-1)^m$ the sign*

of the spinor ±i, is the isomorphism $\mathbb{Z}/6 \cong C_3 \times C_2$. Each branch meets each spinor sign exactly once per cycle.

Proof. The frames with $k \in \{2, 6\}$ are those with $n \equiv 2 \pmod 4$. For $n = 4m + 2$, $j \equiv m + 2 \pmod 3$ and $k = 2$ or 6 as m is even or odd, so the spinor is $+i$ or $-i$. The pair $(m \bmod 3, m \bmod 2)$ determines $m \bmod 6$ by the Chinese remainder theorem. ■

The orbital calculator has six channels with the symmetry $C_3 \times C_2$ (Hoard 2026a), and the cycle produces six handshakes with the same symmetry, in a fixed cyclic order. That the calculator's channels are these handshakes is a proposal, not a theorem.

Table 12 states, for each construction of this paper, what the Engine supplies and whether its method is the most accurate available.

Table 12: What the Engine supplies, construction by construction.

Construction	What the Engine supplies	Most accurate method?	Status
Spinor clock	The Gaussian generator 1 + i with gcd reduction, in two signed bits	Yes: exact, while e^{ikπ/4} in floating point is not, and no rational clock of order 8 exists (Proposition 2.4)	**[PROVEN]**
Nine-phase wheel and tetrahedra	The circle-to-sphere lift of Lemma 8	Exact for the cube triad; the triads at 40° and 80° have irrational coordinates (cos 40° has degree 3), so the Engine is their source, not an exact method	**[PROVEN]**
Conformal lift and null cone	Its use for the Engine's distance constraints (axiom C2)	The method is standard conformal geometry; the Engine is the source of its use	**[ESTABLISHED]**
Joint schedule	The register convention k = n mod 8, j = n mod 3	Exact by the Chinese remainder theorem	**[DEFINITION]**, **[PROVEN]**
Pump holonomy	Block telescoping with exact rational quaternions	Yes: exact, with the open product as the only meaningful test	**[PROVEN]**
{4,3,5} boosts	Golden rapidity, exact in $\mathbb{Z}[\sqrt{\varphi}]$	Yes: exact algebraic integers, against rounding in floating point	**[PROVEN]**
Constant-time evaluation	Projective reduction of finite-order motions	Optimal in the sense of Theorem 5.1: no exact method is constant-time for infinite-order motions	**[PROVEN]**
Mixing of the proposer	Ramanujan quotients of the norm-5 tree	The graphs are Lubotzky, Phillips and Sarnak's; the Engine supplies their use	**[ESTABLISHED]**, **[COMPUTED]**
P versus NP	The reformulation of Theorem 5.11	No method: the problem is open	**[OPEN]**

The pattern is the series' Dual-Helix principle in a new place. Geometry proposes the structures: the wheel, the tetrahedra, the hyperbolic cells, the schedule. Exact arithmetic certifies what they deliver: the integer clock, the rational holonomy, the algebraic boosts. Where the geometry would need irrational or unbounded numbers, the Engine keeps only what exact arithmetic can certify.

8 Status of Claims

Table 13 gives the status of 50 claims: those of the three source texts that this paper examines and those it adds. Abbreviations: TM, Traversal Monograph; ZPA, Zero-Point Addendum; VB, Volumetric Breath; HH, Hyperbolic Horizon; FL, Fifteen Lemmas.

Table 13: Consolidated status of claims.

#	Claim	Source	Status	Where
1	The nine phases are three 120° triads offset by 40°	ZPA §2; FL Lemma 8	**[PROVEN]**	Proposition 2.1
2	Each triad and the apex form a regular tetrahedron of side $2\sqrt{2}$	ZPA §3	**[PROVEN]**	Proposition 3.1
3	In the frame $u = (1, -1, 0)/\sqrt{2}$, the triad $a = 0$ lies on cube vertices	ZPA §3	**[SUPERSEDED]**; true in the corrected frame	Proposition 3.1
4	The tetrahedron and the origin lift to a 4-simplex	ZPA §3.1; FL Lemma 8.3	**[PROVEN]**	Proposition 3.2
5	Five null vectors span R^5 and linearize squared distance	ZPA §3.2; FL Lemma 8	**[PROVEN]**	Proposition 3.3
6	A 2D-to-5D lift creates five intrinsic degrees of freedom	VB (implicit); rejected in ZPA §1	**[SUPERSEDED]**: ambient dimension only	Definition 1.1
7	The C_3 split recovers three channels; simultaneous phasors do not	ZPA §4	**[PROVEN]**	Proposition 3.4
8	The apex stores a parity value	ZPA §4	**[HEURISTIC]**	Section 3.3
9	$(1 + i)^4 = -4$ and $(1 + i)^8 = 16$; states in two signed bits	TM §5.1; FL Lemma 13	**[PROVEN]**	Proposition 2.2
10	A 45° spinor step is a 90° spatial step; eight steps make 720°	ZPA §5	**[PROVEN]**	Proposition 2.3
11	Eight 45° steps make one 360° spatial breath, and spinor closure needs sixteen	VB	**[SUPERSEDED]**	Section 9.2
12	Step 4 is the apex of the breath	VB	**[SUPERSEDED]**: a node of the envelope	Proposition 2.3
13	Steps 2 and 6 give the same rotation with opposite spin	TM §5.1; FL Theorem 7.5	**[PROVEN]**	Proposition 2.3
14	No rational unit quaternion has order 8	New	**[PROVEN]**	Proposition 2.4
15	The joint period is 24 and the same-branch delay 12	ZPA §6	**[PROVEN]** under Definition 3.6	Proposition 3.7
16	The schedule and three clocks close together at 120 frames	New	**[PROVEN]**	Proposition 3.8
17	$24 = \lvert 2T \rvert$, $120 = \lvert 2I \rvert$, $60 = \lvert I \rvert$	New	**[CORRESPONDENCE]**	Section 3.5
18	$720 = 30 \times 24 = 90 \times 8$	ZPA §8	**[PROVEN]**	Proposition 3.7
19	Ten-frame blocks telescope; H = P^{−1}	Hoard 2026b; FL Lemma 12	**[PROVEN]**	Proposition 3.9
20	A return to the identity at frame 720 purges drift	VB	**[SUPERSEDED]**	Proposition 3.13

#	Claim	Source	Status	Where
21	One erased branch is recovered; histories are not	ZPA §7; FL Lemma 7	**[PROVEN]**	Proposition 3.10
22	D_3 folding saves 66.6 %, a fundamental domain of one third	VB	**[SUPERSEDED]**: one sixth for tables; data only if symmetric	Proposition 3.11
23	Every Hurwitz quaternion has an inverse	VB	**[SUPERSEDED]**: in the rational quaternions; only 24 units in the order	Section 9.2
24	Quaternion rotations map from SO(3) into SL(2, $\mathbb{C}$)	VB	**[SUPERSEDED]**: SU(2) $\subset$ SL(2, $\mathbb{C}$)	Section 4.2
25	Hyperbolic projection compresses volume into a finite disk	VB	**[SUPERSEDED]**: volume grows exponentially	Proposition 4.8
26	Maximum diffusion of the sieve at Step 4; a Planck-second; Tesla-valve loops	VB	**[HEURISTIC]**, no derivation	Section 9.2
27	The zero point is an infinitely dense singularity	VB	**[HEURISTIC]**; replaced by a closure state	Definition 3.12
28	No hyperbolic motion has finite order	New	**[PROVEN]**	Proposition 4.1
29	Rotations and boosts are one complex rapidity	New in this form	**[PROVEN]**, **[ESTABLISHED]**	Proposition 4.2
30	The {4,3,5} cell has dihedral angle 72° and cosh $\vartheta = \varphi$	TM §2; FL Lemma 8	**[PROVEN]**	Proposition 4.3
31	The spinor half-rapidity equals the inradius 0.5306	New	**[PROVEN]**	Proposition 4.3
32	1 + i is a hyperbolic screw whose translation is its bit growth	New	**[PROVEN]**	Proposition 4.4
33	The scale register is a boost	New	**[PROVEN]**	Proposition 4.5
34	The cell boundary has curvature 4π; closed {4,3,5} manifolds have $\chi = 0$	New	**[PROVEN]**	Propositions 4.6, 4.7
35	$N_{cubic}(L) = (4/3)L^3 + 2L^2 + (8/3)L + 1$ counts the cubic shadow	TM §6.1	**[SUPERSEDED]**: that is the full ℓ^1 ball	Proposition 4.8
36	The Euler evaluation dichotomy	New	**[PROVEN]**	Theorem 5.1
37	Under C1 to C5, P_GET = NP_GET = TIME_GET(O(1))	TM Theorem 3.3	**[PROVEN]** as restated; true by construction	Theorem 5.6; Proposition 5.7
38	Every procedural O(N) trajectory reduces to three O(1) lookups	HH Theorem 3	**[SUPERSEDED]** in general; **[PROVEN]** for finite order or given tables	Section 5.4
39	P_GET is the class P	TM (implicit)	**[SUPERSEDED]**: a proper subclass	Proposition 5.9
40	P = NP if and only if SAT has an integrable encoding	New	**[PROVEN]**	Theorem 5.11
41	P versus NP	—	**[OPEN]**	Section 5.6
42	The obstructions are strict lower bounds outside C_GET	TM §4	**[SUPERSEDED]**: theorems about their objects	Theorem 5.14

#	Claim	Source	Status	Where
43	Lookup arguments cannot settle P versus NP	New	**[PROVEN]**	Proposition 5.15
44	At most 60 instructions per synthesis	TM §5.2	**[HEURISTIC / COMPUTED]**, Form 1 path	Section 5.10
45	6.66×10^{37} addresses; a compression ratio of 5.16×10^{32}	TM §6.1	Count **[COMPUTED]**; ratio 6.0×10^{32}; not compression	Section 6.1
46	Mixing 4.95 against 318 at 1.1×10^5 nodes; X(5, 101) of order 1,030,200	TM §6.3	**[SUPERSEDED]**: 4.20 against 269.30; order 515,100	Section 6.3
47	Clock windows of 2, 6, 10, 30, 60 and 720 pulses	TM §6.5; FL Lemma 11	**[PROVEN]**	Section 6.4
48	25,920 pulses as a Platonic Great Year	TM §6.5	**[CORRESPONDENCE]**	Section 6.4
49	The six handshakes are labelled by $C_3 \times C_2$	New	**[PROVEN]**; channel identification **[CORRESPONDENCE]**	Proposition 7.1
50	The 24-frame schedule raises throughput	ZPA §6 (as a proposal)	**[CONJECTURAL]**	Problem 3

9 Errata of the Source Texts

The three source texts differ in reliability. The Zero-Point Addendum is sound apart from one frame. The Traversal Monograph proves most of what it states but mislabels its central theorem and miscomputes several table entries. The Volumetric Breath is an informal narrative whose named stages this paper keeps and whose mechanisms it mostly replaces.

9.1 The Zero-Point Addendum

Table 14: Errata of the Zero-Point Addendum.

#	Location	The addendum states	Finding	Resolution
1	§3, after (3.7)	With u = (1, −1, 0)/√2, v = (1, 1, −2)/√6, only the a = 0 orbit coincides with three cube vertices	In that frame the cube vertices of the circle sit at 30°, 150° and 270°, so no wheel point is a cube vertex	Frame u = (2, −1, −1)/√6, v = (0, 1, −1)/√2 (Proposition 3.1)
2	§§3–8	Tags [DERIVED], [PROPOSED SCHEDULING CONVENTION], [PROPOSED ENGINE INTERPRETATION]	Not series tags	Read as **[PROVEN]**, **[DEFINITION]** and **[HEURISTIC]**
3	§10	"Regular tetrahedron: derived, numerically checked"	Holds exactly for every phase	Symbolic proof (Proposition 3.1; Appendix A, section E)

Everything else in the addendum is confirmed here, including its correction of the Volumetric Breath's angle convention, its single-erasure identity, its warning that telescoping keeps only endpoints, and its statement that the lifts do not create physical dimensions or prove P = NP.

9.2 The Volumetric Breath

Table 15: Errata of the Volumetric Breath.

#	Location	The text states	Finding	Resolution
1	Introduction	Eight steps of 45° make one 2π spatial breath; a second breath completes the 4π spinor cycle	With the exact generator 1 + i, each 45° spinor step turns space by 90°: four steps make 360° and eight make 720°	Proposition 2.3; Table 4
2	Step 8; closing paragraph	Iteration 720 is 4π	720 is a frame count; under the exact convention 4π is reached every 8 frames, 90 times in 720	Proposition 3.7(4)
3	Step 4	"The Apex Horizon", the top of the breath	The envelope $\|\sin(k\pi/4)\|$ vanishes at Step 4; the apices are Steps 2 and 6	Proposition 2.3(3)
4	Step 1	Group closure traps the 4 × 4 × 4 tensor in its manifold	Closure under multiplication does not bound norms; integer products grow (Lemma 13 Part 3)	Theorem 5.1
5	Step 3	Rotations map from SO(3) into SL(2, ℂ), the double cover of the Lorentz group	No homomorphism $SO(3) \to SL(2, \mathbb{C})$ splits the cover; quaternions enter as $SU(2) \subset SL(2, \mathbb{C})$	Section 4.2
6	Step 3	Projection compresses the volume into a finite region akin to a Poincaré disk	The Poincaré model draws an infinite volume in a finite picture; hyperbolic volume grows like e^{2R}	Proposition 4.8
7	Step 4	The prime sieve reaches maximum diffusion; momentum halts for a Planck-second	No derivation; the sieve's diffusion is a separate Markov model with no physical time scale	**[HEURISTIC]**
8	Step 5	Every Hurwitz integer quaternion has a perfect inverse	Inverses exist in the rational quaternions; inside the Hurwitz order only the 24 units are invertible	Fifteen Lemmas, Lemma 5
9	Step 5	Collapse routed through Tesla-valve loops	An analogy; the exact counterpart is a non-gradient drift with a cycle current, which circulates without selecting (Hoard 2026d, Proposition 4.8)	**[HEURISTIC]**; Theorem 5.14(V)
10	Step 6	D_3 symmetry makes 66.6 % redundant; the fundamental domain is exactly one third	D_3 gives one sixth (a saving of five sixths) for curve tables, C_3 one third; data folds only if symmetric	Proposition 3.11
11	Step 7	O(1) lookup by static 120° and 240° matrices, bypassing floating point	Correct for the tables, with the knot's axis along (1, 1, 1) so that the matrices are signed permutations	Proposition 3.11
12	Closing paragraph	At iteration 720 the product of all deltas returns violently to the identity and purges floating-point drift	The cyclic product is identically 1 whatever the drift; drift is removed by exact arithmetic, and the test is the open product	Proposition 3.13
13	Introduction	The zero point is an infinitely dense singularity whose energy unspools	No physical content is derived	Definition 3.12

9.3 The Traversal Monograph

Table 16: Errata of the Traversal Monograph.

#	Location	The monograph states	Finding	Resolution
1	Abstract; Theorem 3.3; §7	P_GET = NP_GET = TIME_GET(O(1)) under C1 to C5 is a rigorously proven conditional equivalence **[PROVEN]**	Proven as an implication for problems whose witness is a procedural evaluation; true by construction; constant time only for finite-order motions or bounded horizons	Theorem 5.6; Propositions 5.7 to 5.9
2	Definition 3.1	P_GET and NP_GET as conditional versions of P and NP	TIME_GET(O(1)) is a proper subclass of P; NP-complete problems on unbounded inputs are not in it	Proposition 5.9; Theorem 5.11 for the classical form
3	§2	No cost model is stated	O(1) is undefined without one	Definition 5.3
4	§1.1; Theorem 3.3 part 1	Any O(N) trajectory with shared procedural parameters decomposes into three O(1) lookups (HH, Theorem 3)	True for finite-order generators or when tables hold the powers; false for infinite-order or hyperbolic generators over unbounded N	Theorem 5.1
5	Abstract; §4	Five obstructions are strict information-theoretic lower bounds outside C_GET	Each is a theorem about its own object; none is a lower bound for an NP-complete problem; two are stated for one application only	Theorem 5.14, in general form
6	§5.1	The rays $s \in \{0, 3, 6\}$ end on cube vertices	Frame-dependent	Proposition 3.1(4)
7	§5.1, Step 6	The trefoil scaled by 1/12 has largest Klein radius 0.4706	Its largest Klein radius is 0.2520; the conclusion, that it lies inside the cell of half-width 0.4859, holds	Section 4.3
8	§5.2	At most 60 instructions end to end	Holds on the Form 1 bitmap path; Miller–Rabin certification costs hundreds of operations	Section 5.10
9	§6.1	$N_{cubic}(L) = (4/3)L^3 + 2L^2 + (8/3)L + 1$, with 1,561, 4,353, 16,768 and 1.291×10^5 at L = 12, 17, 27, 54	The formula counts the full ℓ^1 ball (215,929 at L = 54); the column counts the reachable shadow correctly up to L = 8 but not beyond: 1,469, 3,894, 14,644 and 110,935	Proposition 4.8; Table 8
10	§6.1, Prediction 1	A spatial address-compression ratio of 5.16×10^{32}	A ratio of two counts, 6.0×10^{32} with the corrected shadow; not a compression	Section 6.1
11	§6.1, row L = 8	The physical cubit horizon	$5^{10} \le 2.29 \times 10^7 < 5^{11}$: the cubit horizon is depth 10	Table 8
12	§6.3	X(5, 101) has 1,030,200 nodes and X(5, 1009) about 1.03×10^9	5 is a square modulo 101 and 1009, so both are on PSL: 515,100 and 513,621,360	Section 6.3
13	§6.3	Predicted mixing 4.95 against 318 at about 1.1×10^5 nodes	Computed 4.20 against 269.30; the stated bound $\ln(n/2\varepsilon)/(6 - 2\sqrt{5})$ is an upper bound (7.6 for X(5, 13), where the computed time is 3.25)	Table 10

10 Open Problems

Six problems remain open. Each is stated in the manner of a registered report, as in the cubit installment (Hoard 2026d, Section 8): the evidence so far, a test with its decision rule fixed

before any new computation, and whether the results of this paper depend on the answer. They depend on none of them.

Table 17: Open problems, their tests and their bearing on this paper.

#	Problem	Status	Test	Results depend on it?
1	An encoder and decoder for the three branches	**[OPEN]**	Bit-exact round trips with one branch erased	No
2	Integrable encodings for structured classes of SAT	**[OPEN]**	A stated encoding with a proof	No
3	A throughput gain from the 24-frame schedule	**[CONJECTURAL]**	Preregistered simulation against round-robin	No
4	A small closed manifold tiled by {4,3,5} cubes	**[OPEN]**	Construction of a torsion-free subgroup	No
5	Mixing on X(5, 1009)	**[OPEN]**	Direct computation	No; it tests the extrapolation of Table 10
6	P versus NP	**[OPEN]**	None available	No

Problem 1 (Branch Encoder and Decoder). **[OPEN]** Raised by the Zero-Point Addendum, §11.

Specify an encoder E_j for each branch, a reversible offload decoder D_j, a bit-level register map into the Einstein Tile, and a proof of the information each preserves.

Evidence. Proposition 3.10 gives the linear identity that any such scheme must respect: one erased branch is recoverable, two are not, and the average alone determines nothing.

Test and decision rule. Run 10^4 random branch records through 10^3 joint cycles and through 720-frame cycles with non-identity residuals, deleting one branch at random in each. One record not reconstructed bit-exactly refutes the implementation; a claimed recovery from two erasures contradicts Proposition 3.10 and refutes the scheme's proof.

Dependence. None.

Problem 2 (Integrable Encodings of Structured SAT). **[OPEN]** Raised by Theorem 5.11.

Which classes of satisfiability instances admit integrable encodings built from the Engine's constant-time mechanisms, finite-order residues and telescoping?

Evidence. 2-SAT, Horn-SAT and XOR-SAT are solvable in polynomial time **[ESTABLISHED]**, and XOR-SAT is linear algebra over $\mathbb{F}_2$, an abelian and finite-order structure of exactly the kind Corollary 5.2 describes. No class beyond the known tractable ones has been given an encoding.

Test and decision rule. A proposal must state the class, enc and eval, and prove V(x, eval(enc(x))) = 1 on every satisfiable member. One satisfiable instance of the class on which the encoding fails refutes it. An encoding for all of SAT would prove P = NP and would be checked as such.

Dependence. None.

Problem 3 (Throughput of the Schedule). **[CONJECTURAL]** Raised by the Zero-Point Addendum, §6.

Does the 24-frame schedule of Definition 3.6 move more branch data per frame than a round-robin schedule with the same handshake windows?

Evidence. None. Proposition 3.7 shows the schedule is exact and fair; it says nothing about throughput, and its 30 onloads and 30 offloads per branch per 720 frames are event counts, not capacity.

Test and decision rule. In a preregistered simulation with a fixed per-frame transfer budget, measure completed branch round trips per 720 frames under both schedules over 10^3 seeds. The conjecture is supported if the 95 % confidence interval of the ratio lies above 1 and refuted if it lies at or below 1.

Dependence. None.

Problem 4 (A Small Closed {4,3,5} Manifold). [OPEN] Raised by Proposition 4.7.

Find a torsion-free subgroup of small index in the Coxeter group [4,3,5] and the number C of cubes in its quotient.

Evidence. Selberg's lemma guarantees such subgroups; Proposition 4.7 requires 5 | C.

Test and decision rule. Construct congruence quotients of [4,3,5] over the ring of integers of $\mathbb{Q}(\sqrt{5})$, test their kernels for torsion, and report the smallest C found. The result is a construction; a value of C that is not a multiple of 5 would refute the computation, not the proposition.

Dependence. None. A match between C and an Engine count such as $W_\Sigma = 60$ would be a **[CORRESPONDENCE]** only.

Problem 5 (Mixing at Half a Billion Nodes). [OPEN] Raised by Table 10.

What is the 1 % mixing time of the walk on X(5, 1009), with 513,621,360 nodes?

Evidence. The computed times 3.25, 3.80, 4.20 and 4.55 grow by about 0.24 per unit of ln n, which extrapolates to about 6.2.

Test and decision rule. Compute the time on the grid of Table 10, using the vertex-transitivity of the Cayley graph to start from one vertex. A value outside [5.5, 7.0] refutes the extrapolation law, which is then replaced by the fitted law of all five points.

Dependence. None; only the last row of Table 10 uses the extrapolation.

Problem 6 (P Versus NP). [OPEN] Clay Mathematics Institute (2000).

Is P equal to NP?

Evidence. Theorem 5.11 reduces the question to the existence of an integrable encoding for SAT; Proposition 5.15 shows that arguments made only of lookups cannot settle it.

Test and decision rule. None is available. A claimed resolution must be a proof, and it must escape relativization, and, for $P \neq NP$, natural proofs and algebrization.

Dependence. None. Every result of this paper holds whichever way the problem is settled.

11 Conclusion

The zero-point cycle is Euler's formula made exact. Its seed is the nine-phase wheel, three 120° triads that carry three regular tetrahedra with a common apex, a 4-simplex on the paraboloid and five null vectors in R^{4,1}. Its clock is the Gaussian integer 1 + i: a 45° spinor step and a 90° spatial step, −1 after four steps and +1 after eight, held in two signed bits because Niven's theorem forbids a rational clock of order 8. A 24-frame register binds the clock to the branches, with handshakes 12 frames apart, six handshakes labelled by $C_3 \times C_2$, a joint closure with the Engine's clocks at 120 frames, and six such closures in the 720-frame gear. The zero point is the state in which the spinor, the geometry and the memory all pass their tests, and the open holonomy, not the cyclic one, is the memory test.

In hyperbolic space Euler's formula loses its kernel. Rotation and rapidity become the two parts of one complex rapidity; the spinor clock is its elliptic case, the scale register and the golden boosts of the {4,3,5} honeycomb its hyperbolic case, and 1 + i itself a screw whose translation is exactly the half bit it gains per step. The honeycomb's cell has dihedral angle 72°, half-rapidity equal to its inradius, boundary curvature 4π, and closed quotients of Euler characteristic 0.

The complexity statement follows from one dichotomy: an N-step motion is evaluated exactly in constant time if and only if it has finite order. The Engine's conditional collapse of P_GET and NP_GET is therefore true and structural: where the answer is a procedural function of the question, search is evaluation. Constant time holds for cycles and inside fixed horizons, where it holds for every problem, and TIME_GET(O(1)) is a proper part of P. The classical form is exact: P = NP if and only if satisfiability has a polynomial-time integrable encoding. That condition is the P versus NP problem, which remains open, and arguments made of lookups cannot settle it. What the Engine contributes is a precise account of which of its motions close, which grow, and what any proof of P = NP by its methods would have to build.

Appendix A: Computational Verification

The script verify_zero_point.py reproduces every number of this paper and checks every exact identity, by exact integer, rational or symbolic arithmetic wherever the claim is exact and in floating point only for values tagged **[COMPUTED]**. Every check is an assertion, and the run stops at the first failure. It needs Python 3 with numpy, scipy, sympy and mpmath and runs in about 50 seconds on two cores; the option --quick skips the largest mixing computation. All checks pass.

Table 18: Sections of the verification script.

Section	What it checks	Where used
A	μ_9 as three cosets of μ_3; triad sums; e^{iπ} + 1 = 0	Proposition 2.1
B	Powers and reduced states of 1 + i; the integer rotations Rot_x(90k°); the envelope; the SU(2) form	Propositions 2.2, 2.3
C	Finite orders of the 39,624 rational unit quaternions with denominator at most 25; 2T and 2O and their orders	Proposition 2.4
D	The faithful character of $\mathbb{Z}/8$	Proposition 2.5

Section	What it checks	Where used
E	The tetrahedra for every phase, symbolically, in both frames; the cube vertices at 30°, 150°, 270° in the addendum's frame; the 40° rotation	Proposition 3.1
F	Ranks 4 and 5 of the lifts; the null-cone identities; the signature	Propositions 3.2, 3.3
G	The C_3 recovery formulas and the cyclic shift	Proposition 3.4
H	The CRT register, the handshake frames, the delay 12, the six handshakes as $C_3 \times C_2$, the clock closures at 2, 6, 10, 30 and 120	Propositions 3.7, 3.8, 7.1
I	Block telescoping, the partition, Q^{(8)} = Q^{(10)} = P^{−1}, H = P^{−1} on 720 random rational quaternions; the vacuous cyclic product; 13,824 chains to 24 products	Propositions 3.9, 3.10, 3.13
J	Single-erasure recovery; the kernel of the averaging map	Proposition 3.10
K	The trefoil's D_3 and its fundamental arc; g_3 and g_2 generate D_3	Proposition 3.11
L	The hyperbolic Euler formula; the split-complex units; complex rapidity in SL(2, $\mathbb{C}$)	Propositions 4.1, 4.2
M	The golden cell: $a^2 = \sqrt{5} - 2$, the dihedral angle, $\cosh \vartheta = \varphi$, $\tanh(\vartheta/2) = a$, the minimal polynomial of $\sqrt{\varphi}$, the face angle, the cell volume; the trefoil inside the cell	Proposition 4.3; Section 4.3
N	Gauss–Bonnet on the cell; $\chi = 0$ and $5 \mid C$; the hyperbolic ball volume	Propositions 4.6, 4.7, 4.8(1)
O	The Poincaré extension of $z \mapsto (1 + i)z$ is an isometry; translation lengths	Proposition 4.4
P	Dilation as a boost in R^{4,1}	Proposition 4.5
Q	Finite-order powers; denominators 5^N; floating-point error; bit growth of boosts	Theorem 5.1
R	Tree words and their distinct endpoints for $L \le 8$; the cubic shadow by enumeration and dynamic programming for $L \le 20$; Table 8	Proposition 4.8; Table 8
S	Black-box search against an adversary	Proposition 5.13
T	Search-to-decision on random 3-SAT	Theorem 5.11
U	Orders of X(5, q); mixing times on X(5, q) and on cubic tori	Table 10
V	The clock windows of Table 11	Table 11
W	The tier spinor h; parity recovery; the CRT modulus	Theorem 5.6
X	The depths of the precision horizons	Tables 8 and 9

Appendix B: Notation

Table 19: Symbols used in this paper.

Symbol	Meaning
μ_n, ζ_9, ω	The n-th roots of unity; e^{2πi/9}; e^{2πi/3}
$\theta_{a,j}$	Phase $2\pi a/9 + 2\pi j/3$ of family a and branch j
q_k	Spinor state e^{ikπ/4}, held as the reduced Gaussian state of (1 + i)^k
A_k	Transverse envelope $\lvert\sin(k\pi/4)\rvert$
K, V	Core cube $[-1, 1]^3$ and voxel cube $[-2, 2]^3$, in units of C_0
O, 2T, 2O, 2I	Cube rotation group; binary tetrahedral group (the 24 Hurwitz units); binary octahedral group; binary icosahedral group

Symbol	Meaning
c, r, u, v	Centre −(1, 1, 1)/3, radius √(8/3) and orthonormal frame of the phase circle
x*, Δ_a	Apex (1, 1, 1); the tetrahedron of family a
P(x), X(x)	Paraboloid lift (x, ‖x‖²); conformal lift (x, ‖x‖²/2, 1) = x + ½‖x‖²e_∞ + e₀
⟨·, ·⟩	Form of signature (4, 1): x·y − αβ′ − βα′
s, z	Fixed and rotating parts of the C_3 Fourier split
k(n), j(n)	Joint register n mod 8 and n mod 3
Δq_n, ΔQ_m, δ_m	Fine increment, ten-frame block product, middle inverse aggregate
A_ℓ	The ℓ-th block of middle frames, ⌊10m/72⌋ = ℓ
P, H	Open product $q_{719}\, q_0^{-1}$; pump holonomy Q^{(8)}Q^{(10)}P = P^{−1}
j (Section 4)	Split-complex unit, $j^2 = +1$
ϑ, λ	Rapidity; complex rapidity ϑ + iθ
φ, a, r_in	Golden ratio; Klein half-width φ^{−3/2} of the {4,3,5} cell; its inradius artanh a
B, D_λ	Face-neighbour boost; dilation boost diag(1, 1, 1, λ, 1/λ)
α	Face angle arccos(1/√5) of the {4,3,5} cell
S_5	Norm-5 generators 1 ± 2i, 1 ± 2j, 1 ± 2k
N_tree, N_shadow	Tree addresses 6 · 5^{L−1} at depth L; their image in $\mathbb{Z}^3$
X(5, q)	Lubotzky–Phillips–Sarnak Ramanujan graph
h	Tier spinor (1 − i + j + k)/2
C1 to C5, C_GET	Integrability axioms (Definition 5.5)
SEQ(N), NP_GET, P_GET	Classes of Definition 5.4
enc, eval	Integrable encoding (Definition 5.10)
C_0	Acoustic cubit, about 53.748 cm (Hoard 2026d)

References

1. Aaronson, S. & Wigderson, A. (2009). Algebrization: a new barrier in complexity theory. ACM Transactions on Computation Theory 1, article 2.
2. Baker, T., Gill, J. & Solovay, R. (1975). Relativizations of the P =? NP question. SIAM Journal on Computing 4, 431–442.
3. Bennett, C. H., Bernstein, E., Brassard, G. & Vazirani, U. (1997). Strengths and weaknesses of quantum computing. SIAM Journal on Computing 26, 1510–1523.
4. Clay Mathematics Institute (2000). The Millennium Prize Problems: P vs NP. Official problem description by S. Cook.
5. Conway, J. H. & Smith, D. A. (2003). On Quaternions and Octonions. A K Peters.
6. Cook, S. A. (1971). The complexity of theorem-proving procedures. Proceedings of the Third Annual ACM Symposium on Theory of Computing, 151–158.
7. Coxeter, H. S. M. (1973). Regular Polytopes, 3rd edition. Dover.
8. Garner, H. L. (1959). The residue number system. IRE Transactions on Electronic Computers EC-8, 140–147.

9. Hoard, G. (2026). The Fifteen Lemmas of the Geometric Engine: A Unified Projective, Quaternionic, and Metrological Monograph for the SHD-CCP 64-Bit Cubic Protocol. BioChain AI — SHD-CCP Architecture Research, Consolidated Technical Monograph, October 16, 2026. Cited as Fifteen Lemmas.
10. Hoard, G. (2026). The Geometric Engine of Time: Rigorous Formulation of Conditional P = NP, O(1) Hyperbolic Traversal Analysis, and Multi-Scale Predictions. Technical monograph, October 2026. Cited as Traversal Monograph.
11. Hoard, G. (2026a). The Geometric Engine of Time: Mathematical Architecture of the 6-Channel, 5-Tier Toroidal Orbital Calculator. BioChain AI — SHD-CCP Architecture Research, Consolidated Technical Report (Continued Series).
12. Hoard, G. (2026b). The Torsional Markov Memory-Offload Pump: A Three-Scale Quaternionic Delta-Chain with Inverse Compensation and Non-Equilibrium Circulation. Owl Academy / BioChain AI Technical Report.
13. Hoard, G. (2026d). The Geometric Engine of Time: A Proposal for a Standardization of the Cubit. BioChain AI — SHD-CCP Architecture Research, Consolidated Technical Report (Continued Series).
14. Hoard, G. (2026e). The Geometric Engine of Time: The Fokker-Planck Prime Sieve in Quaternionic Lattices. BioChain AI — SHD-CCP Architecture Research, Consolidated Technical Report (Continued Series).
15. Hoard, G. (2026h). The Geometric Engine of Time: The Fokker-Planck Prime Sieve in Quaternionic Lattices — Companion: Status of Claims, Audit and Open Problems. BioChain AI — SHD-CCP Architecture Research, Consolidated Technical Report (Continued Series).
16. Johnson, N. W., Kellerhals, R., Ratcliffe, J. G. & Tschantz, S. T. (1999). The size of a hyperbolic Coxeter simplex. Transformation Groups 4, 329–353.
17. Levin, L. A. (1973). Universal sequential search problems. Problems of Information Transmission 9, 265–266.
18. Lubotzky, A., Phillips, R. & Sarnak, P. (1988). Ramanujan graphs. Combinatorica 8, 261–277.
19. Memory in the Twelve-Engine Hierarchy (2026). SHD-CCP research note, 16 October 2026. Cited as Twelve-Engine Memory.
20. Navigating the Hyperbolic Horizon: Achieving O(1) Traversal in O(N) Sequential Systems using 64-bit Quaternion Cubes (2026). SHD-CCP series paper. Cited as Hyperbolic Horizon.
21. Niven, I. (1956). Irrational Numbers. Carus Mathematical Monographs 11. Mathematical Association of America.
22. Ratcliffe, J. G. (2006). Foundations of Hyperbolic Manifolds, 2nd edition. Springer.
23. Razborov, A. A. & Rudich, S. (1997). Natural proofs. Journal of Computer and System Sciences 55, 24–35.
24. Selberg, A. (1960). On discontinuous groups in higher-dimensional symmetric spaces. Contributions to Function Theory, Tata Institute of Fundamental Research, Bombay, 147–164.
25. The Volumetric Breath of the Zero-Point: An 8-Step π/4 Guide to Cubic Expansion, Hyperbolic Projection, and Symmetry Folding (2026). Text generated in a Gemini session, 17 October 2026. Cited as Volumetric Breath.

26. Zero-Point Eight-Step Pump and Three 120-Degree Conformal Lifts (2026). Research addendum prepared for G. Hoard, BioChain AI — SHD-CCP Architecture Research, 27 October 2026. Cited as Zero-Point Addendum.

Submission details. Corresponding author: [name, affiliation, ORCID and email to be completed before submission]. Keywords: Geometric Engine of Time; Fifteen Lemmas; zero point; Euler's formula; spinors; Gaussian integers; Niven's theorem; conformal geometry; hyperbolic geometry; {4,3,5} honeycomb; golden ratio; Gauss–Bonnet; Ramanujan graphs; P versus NP; relativization; Einstein Tile; SHD-CCP.